II श्री गणेश प्रसन्न II

CAMPBELL-WALSH-WEIN · BOOK 5

Paediatric Urology 2

Concise rapid revision · facts answered in Papers 1–4 point to that answer

highlighted number to memorise · KEY POINT and ALGORITHM boxes carry the reasoning · P2 = covered in full in that Paper's answer book; the list under it gives what that answer misses

Chapter	Topic
Ch 32	Embryology of the Human Genitourinary Tract
Ch 33	Urologic Aspects of Pediatric Nephrology
Ch 34	Perinatal Urology
Ch 35	Urologic Evaluation of the Child
Ch 36	Pediatric Urogenital Imaging
Ch 37	Infection and Inflammation of the Paediatric Genitourinary Tract
Ch 38	Core Principles of Perioperative Management in Children
Ch 39	Principles of Laparoscopic and Robotic Surgery in Children
Ch 40	Clinical and Urodynamic Evaluation of Lower Urinary Tract Dysfunction in Children
Ch 42	Bladder Anomalies in Children
Ch 43	Exstrophy–Epispadias Complex
Ch 44	Prune-Belly Syndrome
Ch 45	Posterior Urethral Valves
Ch 46	Neuromuscular Dysfunction of the Lower Urinary Tract in Children
Ch 47	Functional Disorders of the Bladder and Bowel in Children
Ch 48	Lower Urinary Tract Reconstruction in Children
★	Putting it together: common paediatric pathways

32 Embryology of the Human Genitourinary Tract

1 · Overview

- Development is mostly complete by the second trimester. The **indifferent stage** lasts to about **8–9 wk**; then androgens drive the sexes apart.

2 · Kidney and ureter

Bud position decides kidney quality and orifice site.

Three successive kidneys

Kidney	Features
†Pronephros	Cervical, week 4; non-functional and regresses
Mesonephros	†Transient, but its **Wolffian duct** persists; tubules become efferent ductules (male) or epoöphoron (female)
Metanephros (permanent)	†Week 5; ureteric bud and metanephric blastema form the nephrons

Reciprocal induction †

- The **ureteric bud** grows from the Wolffian duct into the metanephric mesenchyme and forms the ureter and collecting system.
- Mesenchymal **GDNF** binds **RET** on the bud to drive branching. **WT1** makes the mesenchyme competent, and loss of **PAX2** causes agenesis or hypoplasia (renal-coloboma syndrome).
- Nephrogenesis ends at about 34–36 wk; no nephrons form after birth.

> **KEY POINT**
> One defect gives both dysplasia and an abnormal orifice: bud position on the Wolffian duct sets where it meets the blastema (kidney quality) and where its orifice lands in the bladder.

Ascent and rotation †

- The kidney ascends in weeks 6–9 and its hilum rotates medially; failure gives a pelvic or malrotated kidney. Fused lower poles make a **horseshoe kidney**, trapped under the inferior mesenteric artery.

3 · Cloaca, bladder, trigone and urethra

P2 Embryogenesis of urinary bladder — urorectal septum, vesicourethral canal vs UGS, wall layers, trigone (apoptosis, mostly bladder muscle), urachus, Weigert–Meyer

Not in the paper answer:

- Anorectal malformations arise from the **cloacal membrane**, not the urorectal septum (weeks 5–6).

Weigert–Meyer rule (duplex kidneys)

P3 Embryology and management of ectopic ureter. — bud too low gives lateral refluxing lower pole; too high gives caudal/medial ectopic upper pole; male vs female ectopic sites and sphincter

Not in the paper answer:

- Male ectopia stays above the external sphincter, so there is no continuous wetting; †female ectopia below it causes continuous dribbling with normal voids.

- With an **ectopic vas**, the vas joins the ureter and both drain into the prostatic urethra.

4 · Gonads

Two testicular hormones explain every disorder of sex development (DSD).

P1 Testicular development, gonadal descent, abnormalities of descent and bilateral undescended testis — indifferent gonad, PGC migration, SRY then Sertoli cords, rete/efferent ductules, AMH/testosterone/INSL3 table, remnants

Not in the paper answer:

- **SRY** makes Sertoli cells enclose primordial germ cells in cords at about **6 wk**; Leydig cells appear from 8 wk.
- **Müllerian-inhibiting substance (MIS, AMH)** from Sertoli cells makes the Müllerian ducts regress at **8–10 wk**; without it, the uterus and tubes persist.
- **Testosterone** from Leydig cells turns the Wolffian duct into epididymis, vas and seminal vesicle.

Ovary — the default

- Without SRY there is no MIS or androgen, so development is female.

> **KEY POINT**
> Two independent testicular signals act on two independent duct systems: MIS decides Müllerian, testosterone decides Wolffian and external genitalia. Every DSD is a combination of the two.

5 · Gonadal descent

P1 Testicular development, gonadal descent, abnormalities of descent and bilateral undescended testis — CSL + gubernaculum; transabdominal (10–15 wk: androgen, INSL3) and inguinoscrotal (7th month: genitofemoral nerve CGRP) phases; female ligaments

6 · Genital ducts

Male accessory glands

- The **prostate** buds at **10 wk** from urogenital sinus (UGS) epithelium, mostly lateral to the **verumontanum**.
- Androgen acts via the **androgen receptor in mesenchyme**; stromal type 2 5α-reductase makes **dihydrotestosterone (DHT), which has about 10× the AR affinity** of testosterone. Its deficiency therefore gives a rudimentary prostate and severe hypospadias.

Female tract

- The Müllerian ducts use the **Wolffian duct as a guidewire**, which explains combined Müllerian and renal anomalies.
- The **entire vaginal epithelium and hymen are UGS-derived** (Bulmer view). Female Wolffian remnants include the **Gartner duct**.

7 · External genitalia

P3 Embryology of the penis and management of proximal hypospadias (including Koyanagi repair) — anogenital distance first, germ layers, double zipper (opening androgen-independent, closing dependent), glanular canalisation, timetable, 12 o'clock nerve-free zone

Not in the paper answer:

- The tunica is thickest at the 12 o'clock nerve-free zone, the dorsal plication site for mild to moderate curvature.
- The shaft is fused by **16 wk**. In females the folds do not fuse, leaving an open vestibule.

8 · Developmental step and resulting anomaly

Step that fails	Anomaly
†No bud or no induction (GDNF–RET, PAX2)	Renal agenesis (absent hemitrigone)
†Abnormal bud–blastema contact	Dysplasia, hypoplasia, MCDK
†Early bifurcation or two buds	Bifid ureter or complete duplication
Bud too low / too high	VUR / ectopic ureter
†Ascent or rotation fails	Pelvic, malrotated or horseshoe kidney
Cloacal membrane or septation	Anorectal malformation; †cloacal anomalies, exstrophy
Allantois persists	Urachal remnants
Fold fusion arrests	Hypospadias
Urethral plate splits	Urethral duplication
Androgen excess in 46,XX	Virilisation (CAH)
Cranial suspensory ligament retained	Intra-abdominal testis
Gubernacular attachment abnormal	Canalicular or ectopic testis
Gubernacular migration fails	Congenital cryptorchidism
Processus fails to close / to involute	Hydrocele, hernia / ascending testis
One Müllerian duct absent	Unicornuate uterus
Ducts fail to fuse	Uterus didelphys
Septum persists	Uterine obstruction or duplication
Vaginal plate fails to canalise	Transverse vaginal septum
Inferior end fails to open	Imperforate hymen, the commonest vaginal obstruction
Müllerian and Wolffian both abnormal	OHVIRA

9 · Clinical correlations

- **Hypospadias** (about 1 in 250) is an **arrest, not a deformation**.
- **CAH** may make a 46,XX infant look fully male. Suspect it with **impalpable gonads and hyperpigmentation**, since it mimics hypospadias with bilateral impalpable testes.
- A **streak gonad** makes no MIS or testosterone, so the child is female; give hormones and **remove the streak if 46,XY**, because of gonadoblastoma risk.
- **Persistent Müllerian duct syndrome** is 46,XY with uterus and tubes, usually from an MIS-receptor defect, found at exploration for bilateral abdominal testes.
- **OHVIRA** presents after menarche with cyclic pain or pyocolpos. The kidney is often **dysplastic with an ectopic ureter into the obstructed hemivagina**; if missed, leakage persists after septum resection.

KEY POINT

Homologues: Littre and Skene; Cowper and Bartholin; spongiosum and vestibular bulb; gubernaculum and the ovarian plus round ligaments; labioscrotal swellings and labia majora. Traps: the vagina is UGS-derived; the appendix testis is Müllerian and the appendix epididymis Wolffian.

33 Urologic Aspects of Pediatric Nephrology

1 · Renal function in the child

- CAKUT is the **commonest cause of paediatric end-stage kidney disease (ESKD)**.
- GFR reaches the adult level by **2 years**. Creatinine in the first 48 h is **maternal**; by day 7 it is the neonate's own (**<0.5 mg/dL** at term).
- †Schwartz: estimated GFR = 0.413 × height (cm) / creatinine (mg/dL).

> **KEY POINT**
> The neonatal kidney is "hormonally deaf": maternal creatinine, salt wasting, hyperkalaemia, low bicarbonate and poor concentration reflect immaturity, not disease.

2 · Haematuria

- Microscopic haematuria is **>5 red blood cells (RBCs)/high-power field**; it affects about 1% of school children and mostly resolves.
- **Fresh microscopy is critical**: red urine is non-glomerular, while brown urine with RBC casts is glomerular. A positive dipstick without RBCs means haemoglobin or myoglobin.
- **UTI is the commonest cause of gross haematuria**. Trauma, pain or clots need urological work-up, often CT; suspected GN needs complement, ASO and nephrology referral.
- In isolated microscopic haematuria, **VCUG and cystoscopy are not indicated**.

3 · Proteinuria

- Normal urine protein-to-creatinine ratio (UPCR) is **<0.2 mg/mg** (<0.7 in newborns).
- **Transient** (fever, exercise) and **orthostatic** proteinuria are benign; **fixed** proteinuria needs full evaluation with or without biopsy.

4 · Glomerular disease

Under 1 year causes are genetic; at 1–10 years think of minimal change, post-streptococcal GN, HSP and HUS; from 7 years, lupus, IgA, MPGN and Alport.

Nephrotic syndrome

- Proteinuria **>40 mg/m²/h or UPCR >2.0** with hypoalbuminaemia and hyperlipidaemia.
- In primary disease give **steroids first, without biopsy**, since minimal change is 90% under 10 years and mostly steroid-sensitive.

Category	Definition	Action
Steroid-sensitive	Remission within 8 weeks	~50% relapse
Frequently relapsing	≥2 relapses in 6 months or ≥4 in 12	Steroid-sparing: cyclophosphamide, mycophenolate, rituximab, calcineurin inhibitors
Steroid-dependent	2 consecutive relapses on steroids	
Steroid-resistant	No remission after 6–8 weeks	Biopsy; usually FSGS, which recurs after transplant

Nephritic syndrome

Complement sorts the causes:

C3	C4	Diagnoses
Low	Normal	Post-infectious GN, C3 glomerulopathy
Low	Low	Lupus nephritis, MPGN
Normal	Normal	IgA, HSP, Alport, ANCA GN, anti-GBM

- In **post-streptococcal GN**, C3 normalises by **8–12 weeks**; if it stays low, biopsy.
- **IgA nephropathy** gives painless gross haematuria **during** a respiratory infection. **Alport** adds sensorineural deafness.
- **HUS** is the commonest cause of severe AKI in a young child; atypical (complement) HUS gets eculizumab.

5 · Tubulopathies

P2 Renal tubular acidosis — types 1/2/4 compared: defect, urine pH, K^+, stones, causes (obstruction gives type 4), treatment

Not in the paper answer:

- Type 4 RTA (**hyperkalaemia**) from obstruction improves after urological correction.

6 · Cystic disease

- **MCDK** has no function, and bilateral disease is lethal. In MCDK, VUR or PUJ obstruction occurs in up to 25%, so do a VCUG, especially with hydroureteronephrosis. Follow with US and BP until involution, then yearly.
- **ARPKD** causes enlarged kidneys and hepatic fibrosis, with >75% ESKD by 5 years. Nephronophthisis causes polyuria and ESKD by adolescence.

7 · Hypertension

- Hypertension is repeated BP **>95th percentile** for age, sex and height.
- Infants have coarctation or renovascular disease, young children Wilms or neuroblastoma, school-age children **renal parenchymal disease including reflux nephropathy**, and obese adolescents essential hypertension.
- Drugs are for stage 2, symptoms, end-organ damage or a secondary cause. Target <90th percentile, or **<50th** in CKD.

8 · Acute kidney injury

KDIGO stage	Creatinine	Urine output
I	Rise ≥0.3 mg/dL in 48 h or 1.5–1.9× baseline	<0.5 mL/kg/h for 6–12 h
II	2.0–2.9×	<0.5 for ≥12 h
III	≥3× or ≥4.0 mg/dL or clearance <35 (<18 years)	<0.5 for ≥24 h or <0.3 for 12 h

- Creatinine lags GFR by 24–72 h. FeNa **<1%** means prerenal and >3% intrinsic.
- Dialysis is for diuretic-resistant overload, refractory hyperkalaemia or acidosis, or uraemia.

9 · Chronic kidney disease

- CKD stages by GFR are I ≥90, II 60–89, III 30–59, IV 15–29 and V **<15** or dialysis.
- Complications include growth failure from growth hormone resistance, bone disease, anaemia and acidosis.
- **Do not restrict protein**; strict BP control is the only proven way to slow progression.

10 · Dialysis and transplantation

P1 Urological evaluation and management of pediatric kidney transplant patient. — lower-tract evaluation and reconstruction before transplant, native nephrectomy indications, ureteric techniques, complications

Not in the paper answer:

- Peritoneal dialysis is first choice if **<9 years / <20 kg**; omphalocele, gastroschisis, diaphragmatic hernia and exstrophy contraindicate it.
- Living donors give less delayed graft function (5% vs 15%) and a longer graft half-life (20 vs 12.5 years).
- Many centres await **≥10 kg and 65 cm** before transplant because of graft thrombosis.

34 Perinatal Urology

1 · Fetal renal development and prenatal imaging

Amniotic fluid is fetal urine, so obstruction threatens the lungs.

P3 Antenatal hydronephrosis (UTD) and fetal uropathy (LUTO) — UTD 1–3% of pregnancies, second after cardiac; 18–20 wk scan; APD, SFU and UTD grading; A1/A2–3 and P1–P3 tables; causes with incidence; postnatal plan

Not in the paper answer:

- The metanephric kidney appears at **day 28** and urine production starts at **14 weeks**. After **~16 weeks** amniotic fluid is mainly urine, so obstruction causes oligohydramnios and pulmonary hypoplasia.
- The amniotic fluid index (AFI) is normally 8–18 cm; oligohydramnios is **<5–6 cm**.
- Ultrasound severity tracks PUJ obstruction and posterior urethral valves (PUV), but not VUR or duplex. Fetal MRI is used only for anatomy, such as cloaca or exstrophy.

> **KEY POINT**
> Prenatal detection is for triage and counselling; there is no proof it improves outcome.

2 · Grading systems

- **Anteroposterior pelvic diameter (APD)** is measured transversely at the hilum and is **meaningless without its gestational age**. Severe is >10 mm in the 2nd trimester and >15 mm in the 3rd.
- Postnatal pathology rises from about 12% (mild) to 88% (severe), but **VUR shows no trend**.
- **SFU** grading (postnatal): 1, sinus barely split; 2, major calyces; 3, minor calyces with parenchyma spared; 4, parenchymal thinning. All SFU 3 and about 70% of SFU 4 kidneys keep DRF >40%.
- Pyramidal thickness **≤3 mm** predicts pyeloplasty and reduced function.
- The **urinary tract dilation (UTD)** classification uses six parameters plus oligohydramnios and covers the whole tract, pre- and postnatally.

3 · Prenatal follow-up

> **ALGORITHM**
> 1. Normal APD ceiling, all else normal: **<4 mm (16–27 weeks) → <7 mm (≥28 weeks) → <10 mm (postnatal >48 h)**.
> 2. If A1 (APD 4 to <7 mm at 16–27 wk, or 7 to <10 mm at ≥28 wk), repeat US once after 32 weeks.
> 3. If A2–3 (any abnormal parameter), scan every 4 weeks until delivery.
> 4. In both, do a postnatal US after 48 h.

- Exstrophy, cloaca, ARPKD, outlet obstruction and myelomeningocele need a multidisciplinary prenatal consult.

4 · Fetal LUTO and oligohydramnios

Fetal intervention saves lungs, not kidneys.

- The picture is a male fetus with bilateral hydroureteronephrosis, a non-cycling bladder, the **"keyhole" sign** and oligohydramnios. Only fetal cystoscopy can separate PUV, urethral atresia and prune-belly syndrome (PBS).
- Prognosis is poor with oligohydramnios **before 27 weeks**, cortical cysts or echogenicity.
- Fetal urine biomarkers are **not validated** (European consensus 2022). †Classic favourable values were Na <100 mEq/L, Cl <90 mEq/L, osmolality <200 mOsm/L and β2-microglobulin <4 mg/L.

Ruano stage	Fluid / kidneys	Urine biochemistry	Intervention
I — mild	Normal fluid and echogenicity	Favourable	Not indicated
II — severe, function preserved	Oligo/anhydramnios; hyperechoic; no cysts	Favourable within 3 samples	**Indicated** — for lungs and kidneys
III — severe, function abnormal	Anhydramnios; hyperechoic, with or without cysts and dysplasia	Unfavourable after 3 samples	Maybe — for lungs only

Options

- Options are early delivery, vesicoamniotic shunt (VAS), fetal cystoscopy and serial amnioinfusion; the window is **18–24 weeks**.
- Complications occur in about **44%**, most often shunt failure, followed by preterm labour and membrane rupture.

Feature	VAS	Fetal cystoscopy
Diagnoses cause	No	Yes
Mechanism	Bladder-to-amniotic bypass	Laser valve ablation
Specific complications	Blockage, migration, bowel herniation	Urethral fistula (≤10%)
Survival / renal function (PUV)	Improved / not improved	Improved / short-term only

- After VAS, over half have renal insufficiency and a third need renal replacement therapy.

> **KEY POINT**
> Counsel: intervention improves perinatal survival (lungs); complications exceed 40%; it does not reliably prevent renal replacement, because dysplasia is often established first.

5 · Postnatal imaging

- Do the **first US after 48 h**, because neonatal dehydration under-reads dilation.
- Postnatal P1 (APD 10 to <15 mm, central calyces), P2 (≥15 mm, peripheral calyces or abnormal ureter) and P3 (abnormal parenchyma or bladder) roughly equal SFU 1–2, 3 and 4. **Classify by the highest-risk feature**.
- **VCUG** finds VUR in up to 31%; order it by **UTI risk** (females, uncircumcised males, high-grade dilation, dilated ureter, abnormal bladder).
- Contrast-enhanced voiding urosonography avoids radiation but shows the urethra poorly.
- **MAG3** is preferred over DTPA. Consider surgery if **DRF <40%**, T½ >20 min or retention persists upright. Pitfalls are dehydration and poorly timed furosemide.
- **DMSA** shows scars, dysplasia and duplex upper-pole function; **MRU** shows ectopic ureters and crossing vessels but needs sedation.

6 · Postnatal management, UTI and prophylaxis

- **P1** needs only a US at 3–6 months; **P2** needs CAP **if the ureter is >7 mm**; **P3** needs CAP, VCUG and MAG3 if dilation persists.
- In P3 without VUR, give CAP until pyeloplasty or **1 year**, as about 90% of UTIs occur in year 1.
- UTI occurs in 8–22%; agreed risks are **female sex, high-grade dilation and intact foreskin**. Circumcision has a number needed to treat of **10**.
- Evidence for CAP is weak. Use amoxicillin or a cephalosporin from birth; nitrofurantoin or trimethoprim only from **≥8 weeks**.

ALGORITHM

1. US after 48 h: if normal, the dilation was transient.
2. If P1, repeat US at 3–6 months.
3. If P2, give CAP if the ureter is >7 mm; VCUG and MAG3 at discretion.
4. If P3, give CAP, do a VCUG and repeat US at about 1 month; if dilation persists, do a MAG3.
5. If bilateral severe dilation, treat as outlet obstruction: decompress, CAP, urgent evaluation.

7 · Specific antenatal diagnoses

- **Transient** dilation (50–70%) is commonest, and mild dilation resolves in 90–98%, mostly by 2–4 years.

PUJ obstruction pattern

- It is the commonest **surgical** cause. Third-trimester APD **15 mm** has about 80% sensitivity and specificity for pyeloplasty.
- Serial US suffices if P2, APD <20 mm and pyramidal thickness >3 mm, since DRF >40% is likely.
- Operate for rising dilation, DRF <40% with poor drainage, falling DRF, upright retention or symptoms.

Vesicoureteral reflux

- Prenatally detected VUR is typically male, high-grade and bilateral, with congenital dysplasia; VUR after febrile UTI is typically female, lower-grade and acquired. Screen girls with high-grade UTD and uncircumcised boys.

Megaureter / ureterovesical junction obstruction

- Most resolve; surgery is predicted by a ureter **>13 mm** and DRF <30%.
- BAPU: a prenatal ureter **>7 mm after 30 weeks** needs postnatal US, CAP and VCUG.
- Operate for DRF <40%, febrile UTI or pain. **Under 1 year, temporise** with ureterostomy or a refluxing reimplant, and reimplant after 1 year.

KEY POINT

In PUJ, VUR and megaureter, dilation alone is not an indication: operate for function loss, infection or progression; grade sets surveillance.

LUTO conditions

- **PUV** is the commonest LUTO. **PBS** combines megacystis, a deficient abdominal wall and impalpable testes.
- **Urethral atresia** means a catheter will not pass; it is managed by vesicostomy, then progressive dilation of the anterior urethra (PADUA). Massive reflux and MMIHS also cause megacystis.

> **KEY POINT**
> Bilateral dilation with normal fluid means urine still leaves the fetus (reflux, megacystis); with low fluid it means true outlet obstruction.

Neonatal management of LUTO

> **ALGORITHM**
> 1. Have the team at delivery and resuscitate the lungs first.
> 2. Pass a 5 Fr feeding tube, with a finger in the rectum to guide it.
> 3. Confirm position on a plain film (KUB).
> 4. If it fails, place a US-guided suprapubic catheter.

- **Do not use a balloon catheter**, which obstructs the posterior urethra and ureteric orifices. Judge decompression by the **creatinine nadir**, and warn of post-obstructive diuresis.
- Oliguria after decompression suggests a thick bladder obstructing the ureters; treat with antimuscarinics.

Multicystic dysplastic kidney

- VUR occurs in about 20%. There is **no Wilms association**, so no routine nephrectomy; follow with US and blood pressure.

Duplex, ureterocele, ectopic ureter

- By **Meyer–Weigert**, the upper-pole orifice is inferomedial and obstructs, while the lower-pole orifice is superolateral and refluxes.
- Suspect a duplex when the ureter is more dilated than the pelvis.
- Endoscopic puncture of a ureterocele is increasingly first-line, and mandatory for outlet obstruction or sepsis. High-grade VUR usually needs ureterocele excision and common-sheath reimplant.

Exstrophy

- Both forms show a non-visualised bladder, low umbilicus and pubic diastasis. **Cloacal exstrophy (OEIS)** adds omphalocele, imperforate anus and spinal anomalies.

Cloaca, urogenital sinus, CAH

- A **cloaca** shows a single channel, a pelvic cyst and calcified meconium; a **urogenital sinus** has no bowel obstruction and normal external genitalia.
- **CAH** (21-hydroxylase, autosomal recessive) is 75% salt-wasting. Only **1 in 8** at-risk pregnancies is an affected girl, so prenatal treatment is controversial.

Myelomeningocele

Fetal closure before 26 weeks reduces shunting and improves motor outcome, but gives **no bladder benefit**.

8 · Neonatal urological emergencies

Urinary ascites and retention

- Urine causes up to 30% of neonatal ascites, and **PUV is the commonest cause**.
- Spontaneous bladder rupture heals with a catheter, but injury from an umbilical catheter needs repair.
- Retention >24 h suggests PUV or atresia in boys, and urogenital sinus or ureterocele in girls.

Abdominal masses

- **Congenital mesoblastic nephroma** is the commonest solid renal mass under 6 months.
- **Adrenal haemorrhage** has no Doppler flow, shrinks on serial US and calcifies like an eggshell, whereas **neuroblastoma** is vascular and persists. Haemorrhage may track down the processus and mimic an acute scrotum.

Anorectal malformation and ambiguous genitalia

- Anorectal malformation has genitourinary or spinal anomalies in up to **75%**, so get renal and spinal US.
- In ambiguous genitalia, defer gender assignment and **exclude salt-wasting first**; the commonest cause is CAH.

Interlabial masses

Mass	Position	Management
Imperforate hymen	Midline bulging membrane	Hymenal incision
Urethral prolapse	Midline collar at meatus	Conservative
Prolapsed ureterocele	Off-midline from meatus; hymen intact	US and VCUG
Paraurethral cyst	Off-midline, displaces meatus	Most resolve

Gross haematuria

First exclude an oestrogen-withdrawal bleed and urate crystals, then get a Doppler US.

Feature	Renal vein thrombosis	Renal artery thrombosis	Cortical necrosis
Risk	Maternal diabetes, prematurity, umbilical venous catheter, polycythaemia	Umbilical artery catheter	Hypoperfusion
Presentation	**Triad**: haematuria, flank mass, thrombocytopenia	Hypertension, heart failure	Renal failure
Treatment	Hydration, with or without anticoagulation	Remove catheter, with or without thrombolysis	Supportive

Scrotal masses and torsion

- **Prenatal torsion** gives a hard, non-tender testis at birth with almost no salvage, so do urgent contralateral orchidopexy.
- **Postnatal torsion** (first 30 days) has salvage up to 44%, so explore immediately and fix the other side.

9 · Landmark trials

Trial	Design	Result	Take-home
PLUTO	RCT, VAS vs conservative in fetal LUTO; 31 randomised, stopped early	Fewer perinatal deaths; no renal benefit; 44% complications	Shunting saves lungs, not kidneys
Great Ormond Street pyeloplasty RCT	APD ≥15 mm, DRF >40%: pyeloplasty vs observation	Most observed kidneys stable or improved; a minority (APD 20–40 mm) lost function	Observation safe with monitoring
MOMS†	RCT, prenatal vs postnatal myelomeningocele closure	Less shunting, better motor outcome; no urodynamic benefit	Fetal closure is not for the bladder

35 Urologic Evaluation of the Child

1 · Triage by acuity

Triage asks: **what is lost by waiting** — an organ, a life, or nothing?

Tier	Conditions
Emergent	Acute scrotum; acute abdomen; AKI; obstruction with UTI and/or AKI; retention; incarcerated hernia; paraphimosis; priapism; trauma; newborn ambiguous genitalia, exstrophy, renal vein thrombosis
Urgent (24 h)	Gross haematuria **without infection**; newborn bilateral or solitary-kidney hydronephrosis; renal or testicular tumour; spina bifida
Semiurgent (1–3 wk)	After treated febrile UTI; postnatal dilation (no outlet obstruction); reducible hernia; interlabial mass; amenorrhoea
Routine	Hydrocele; circumcision; incontinence or enuresis; hypospadias; labial adhesions; meatal stenosis; microscopic haematuria; recurrent UTI; undescended testis; varicocele

KEY POINT

Context shifts the tier: a hernia is emergent if incarcerated but semiurgent if reducible; hydronephrosis is urgent if bilateral or in a solitary kidney.

2 · History

- For hydronephrosis, ask about prenatal history, progression and **amniotic fluid**, a proxy for fetal renal function.
- Obtain **consent from the guardian and assent from the child**; adolescents get a private history.

3 · Examination

Children hide shock until late.

- Minimum systolic BP is **≥60** in neonates, ≥70 at 1 month–1 year, **70 + 2×age** at 1–10 years and ≥90 above 10 years.
- **Shock shows first as tachycardia and tachypnoea; hypotension means decompensation.**

P1 Testicular development, gonadal descent, abnormalities of descent and bilateral undescended testis — examining for undescended testis: warm room, sweep from ASIS, cross-legged/squat, ectopic sites; retractile; no imaging

Not in the paper answer:

- **Micropenis** is a stretched length below mean − 2.5 SD, which is **1.9 cm** at term.
- In **Tanner staging**, the first male sign is testicular enlargement (stage II), not hair.
- A **simple sacral dimple** (<5 mm, within the cleft, <25 mm above the anus) needs no imaging; image for a lipoma, tag, deep dimple or deviated cleft.

4 · Urine and blood tests

- In febrile children under 24 months, use a **catheter or suprapubic aspirate**; **bag specimens have high false-positive rates** and help only if negative.

- The AAP defines UTI as pyuria and/or bacteriuria with **≥50,000 colony-forming units (CFU)/mL** (catheter or aspirate) in a symptomatic child.

Test	Threshold
Microscopic haematuria	**>5 red cells (RBC)/high-power field** (adult ≥3), on >2 occasions
Calcium/creatinine ratio (hypercalciuria)	>0.4 at age 2–5 years; >0.2 above 5 years
Normal protein/creatinine ratio	<0.7 in the first year; <0.4 up to 4 years; <0.2 after 5 years; >3.0 is nephrotic
Specific gravity	<1.010 dilute (water excess, diabetes insipidus); >1.020 dehydration

- Always **confirm dipstick blood by microscopy**, and test proteinuria on a **first-morning** sample.

5 · Imaging

Only US and MRI avoid radiation; keep doses as low as reasonably achievable.

Modality	Answers
US	Kidney, bladder, scrotum, torsion, spine
VCUG	Reflux, urethra, genitogram
CT	Severe trauma, stones
MRI	Tumours, ectopia, duplication, Müllerian or sinus anatomy
MAG3 renogram	**Obstruction** and differential function
DMSA scan	**Scarring** or pyelonephritis, and differential function

6 · Emergent presentations

Acute scrotum

- The commonest cause is **torsion of the appendix testis**, with point tenderness in a prepubertal boy; treat it supportively.
- **Cord torsion** (mostly over 10 years) gives severe pain, vomiting, a high-riding testis and an absent cremasteric reflex.
- It is a **clinical diagnosis**: explore without imaging and fix the other side.

TWIST item	Points	Score and action
Testicular swelling	2	0: no US
Hard testis	2	1–5: scrotal US
Absent cremasteric reflex; nausea or vomiting; high-riding testis	1 each	**≥6**: explore without US

Other emergencies

- In **ambiguous genitalia**, exclude **salt-wasting CAH** (21-hydroxylase in about 95%). Crisis comes on day **4–15**, so do not discharge early; check **17-hydroxyprogesterone and electrolytes**.
- In **exstrophy**, tie the cord with silk (no clamp), cover the plate with cling film and transfer.
- **Infected obstruction** needs prompt drainage by stent or nephrostomy.

- In **paraphimosis**, exclude a **hair tourniquet**, then compress the glans under analgesia; a dorsal slit is the fallback.
- Ischaemic **priapism** is usually from **sickle cell disease**; give oxygen, fluids and analgesia.
- In **trauma**, the **mechanism** decides CT. After straddle injury consider **sexual abuse**; meatal blood needs a retrograde urethrogram, not a blind catheter.
- In **retention**, expected capacity (mL) is **(age + 2) × 30**. Try analgesia and disimpaction before a catheter.

7 · Common referral presentations

Antenatal hydronephrosis

- Scan **before discharge** if bilateral, in a solitary kidney or with suspected outlet obstruction; otherwise after **48 h**.
- A boy with bilateral dilation and a thick bladder needs a **VCUG before discharge**, catheterised meanwhile.

Haematuria

- Glomerular haematuria is tea-coloured with casts and protein; urological haematuria is bright red with clots.
- Investigate all gross haematuria with urinalysis, culture and US. Isolated microhaematuria is usually benign, so repeat it over 2–3 weeks.

Bladder bowel dysfunction and enuresis

- **Always examine the back**, and treat stepwise, starting with behavioural therapy.

Scrotal, inguinal and genital

- A communicating **hydrocele** is smaller in the morning and is treated by high ligation at the internal ring.
- An **undescended testis** with hypospadias or micropenis suggests a DSD.
- For **varicocele**, the Lambert volume is L×W×H×**0.71**; a non-decompressing or right-sided one needs a retroperitoneal US.
- Prepubertal **testicular tumours** are mostly benign, and tumour markers guide testis-sparing surgery.
- **Meatal stenosis** is treated by meatotomy under EMLA, and symptomatic **labial adhesions** by oestrogen or steroid cream.

> **KEY POINT**
> Red flags: absent cremasteric reflex with acute pain; ambiguous genitalia with impalpable gonads; bilateral hydronephrosis in a boy; fever with obstruction; hypotension; abdominal mass; meatal blood; lumbosacral lipoma or deviated cleft.

8 · Child comfort

- Distraction, midazolam, intranasal ketamine and EMLA cream can avoid general anaesthesia.

36 Pediatric Urogenital Imaging

1 · Principles — ALARA and the three harms

- Under **ALARA**, **every child is treated as radiation sensitive**. **Sedation risk usually exceeds contrast risk**, and ultrasonography (US) is **safe at all ages and gestations**.

> **KEY POINT**
> Every choice trades **radiation, sedation and contrast** against yield; the long lifespan tips it towards US, and the question is which test fills US's functional gap (obstruction, reflux, pyelonephritis) most cheaply.

2 · Ultrasonography

US comes first, but shows anatomy, not function.

P1 USG in neonates — technique; neonatal pyramids mimic dilation; under-read in first 48 h; SFU grades 0–4; PUV keyhole, MCDK, ARPKD, masses; normal US does not exclude VUR

Not in the paper answer:

- Fetal urine is made from **10–12 weeks**; by **20 weeks** US detects most anomalies.
- SFU grade 4 is defined by parenchymal thinning, not more dilatation.
- **Bilateral hydroureteronephrosis with a thick, poorly emptying bladder needs a VCUG** for posterior urethral valves (PUV); megaureter or ureterocele needs diuretic renography or MRU.

Cysts, masses, stones

Lesion	Side	US hallmark
MCDK (multicystic dysplastic)	Unilateral	Non-communicating random cysts, scant echogenic parenchyma
ARPKD (recessive polycystic)	Bilateral	Large homogeneously **bright** kidneys, not discrete cysts
ADPKD (dominant)	Bilateral	Parenchyma almost replaced by cysts
Multilocular cystic nephroma	Unilateral	Like MCDK, more parenchyma; infancy
Cystic Wilms	Unilateral	Overlaps; 2–4 years

- A neonatal solid mass is usually **mesoblastic nephroma**; a toddler's is Wilms. Neonatal enlargement with acute kidney injury and hypertension suggests **renal vein thrombosis**.
- **Stones** are echogenic **with** shadowing (fat is without), and Doppler **twinkling artefact** confirms them.

Infection and trauma

- In UTI, US is not routine at the start; **do US if there is failure to improve on antibiotics**. It misses low-grade VUR and scars, where DMSA is better. **Normal US neither risk-stratifies APN nor predicts VUR.**

Contrast-enhanced voiding urosonography (CeVUS)

- Microbubbles grade VUR on five grades with **no radiation**. Area under curve is **0.97**, so it is recommended for VUR.

Bladder/urethra

- The **keyhole sign** (thick bladder and dilated posterior urethra) means PUV.

- To tell clot from tumour, scan supine then prone, because a **clot moves**.

Scrotum and genitalia

- **Never delay exploration for suspected torsion**.
- The **whirlpool sign** gives torsion sensitivity ~96% vs ~76% for colour Doppler, with specificity ~99%.
- Appendage torsion is an avascular nodule with surrounding hyperaemia. **Teratoma** is the commonest prepubertal tumour; epidermoid shows **onion rings**.
- **Do not image a routine undescended testis** (AUA Choosing Wisely) except in an obese child.

3 · Voiding cystourethrography

VCUG is the reference test for reflux and urethral anatomy.

- **After UTI**, do it once the child is improving on antibiotics; a 4–6-week wait is unnecessary, though very early VCUG may overgrade VUR.
- Indications are hydronephrosis with **dilated ureter(s)**, febrile or recurrent UTI, suspected PUV, epididymitis in boys, and UTI in a circumcised boy.
- **Occult VUR**: VCUG finds VUR in only ~40% of DMSA-proven APN. **PIC cystography** (positional instillation of contrast at the orifice) is used when VCUG is negative but febrile UTI recurs.
- **Trauma**: with meatal blood, do a **retrograde urethrogram (RUG) before any catheter**. Capacity (mL) = **(age + 2) × 30**, max 400. A "pie-in-the-sky" bladder means pelvic fracture with extraperitoneal injury.

4 · Nuclear medicine

DMSA

- DMSA is taken up by proximal tubules, so APN shows photon-deficient areas.

Feature	Acute pyelonephritis	Scar
Contour	Preserved	Volume loss, concavity
Timing	Within days; resolves over ~5 months	Lesion at **≥6 months** = scar

- A third to a half of acute lesions scar.

MAG3 diuretic renography

P1 Radionuclide studies and diuretic renography in PUJ obstruction — MAG3 vs DTPA vs DMSA; well-tempered protocol (hydration, catheter, diuretic timing, ROI); DRF, T½ from maximal filling; biphasic curve; pitfalls

Not in the paper answer:

- Give furosemide **1 mg/kg IV** at maximal filling; if given too early, drainage is measured before the system is full.
- There is **no validated paediatric T½ cut-off**, so decide on age, function, curve, upright drainage and serial studies. A **biphasic curve** and poor upright drainage predict progression.

Radionuclide cystography and testicular scan

- **Direct** cystography is more sensitive and lower-dose than VCUG but grades coarsely, so it suits follow-up. **Indirect** cystography is insensitive and not routine.

5 · MR urography

- MRU gives anatomy and function without radiation and suits complex anomalies.
- Protocol: hydration and catheter, then **furosemide 1 mg/kg (max 20 mg)**, T2, gadolinium and dynamic T1; function is measured by **Patlak** analysis.
- It needs sedation under 6 years and is best for **bladder rhabdomyosarcoma** and **local staging of renal tumours**.

6 · CT and radiography

- CT is chosen for speed in trauma and staging, not for hydronephrosis.
- **In trauma, arterial-phase images miss collecting-system injury, so delayed excretory images are mandatory.** CT cystography separates intra- from extraperitoneal rupture.
- CT rarely changes stone care. IVP is obsolete.

7 · Choice of test by clinical question

Question	First line	Next	Avoid
Hydronephrosis — obstruction?	US	MAG3; MRU	CT, IVP
Hydroureteronephrosis, with or without thick bladder	US	VCUG	—
VUR	VCUG	CeVUS; direct cystogram; PIC if occult	Indirect cystogram
APN / scar	DMSA	—	Grey-scale US; MRI
Stones	US	Low-dose CT; KUB	Reflex CT
Renal mass	US	Contrast CT; MRI	—
Bladder mass	US (full bladder)	MRI	—
Trauma	CT with delayed phase; US	RUG; cystography	Catheter before RUG if meatal blood
Acute scrotum	Clinical, with or without Doppler/whirlpool	Scintigraphy if equivocal	Imaging delaying exploration
Undescended testis	Examination	US only if obese	Routine US, MRI
DSD	US and genitography	MRI	CT

37 Infection and Inflammation of the Paediatric Genitourinary Tract

1 · Overview and the paradigm shift

- Urinary tract infection (UTI) affects ~**1% of boys, 3–5% of girls**, and 30–50% recur.
- Congenital dysplasia now explains more scarring than reflux does, so the approach is **selective imaging, prompt treatment and prophylaxis only for the high-risk child**.

2 · Pathogenesis and virulence

- Infection is **ascending**: gut bacteria colonise the perineum, enter the bladder, adhere to umbrella cells and may ascend. Uropathogenic E. coli (UPEC) causes most UTIs.

Factor	Mechanism	Association
Type 1 fimbriae (mannose-sensitive)	FimH binds uroplakins, allowing entry	Cystitis, asymptomatic bacteriuria (ASB)
P fimbriae (mannose-resistant)	Bind Galα1–4Galβ (P blood group)	**Pyelonephritis**
Aerobactin	Siderophore	Growth in iron-poor urine
α-Haemolysin, CNF1	Cell lysis, cytoskeletal damage	Inflammation, invasion
Capsule (K antigen)	Antiphagocytic, anticomplementary	Pyelonephritis
Biofilm, flagella	Shield; motility	Recurrence; ascent

- **Intracellular bacterial communities** (IBCs) form a **quiescent reservoir** that reseeds the lumen for months, causing same-strain recurrence despite adequate antibiotics.

> **KEY POINT**
> IBCs and biofilms shield bacteria from neutrophils and antibiotics; a biofilm survives 1000–1500× normal killing concentrations.

3 · Host risk factors

- In the first year UTI affects boys **3.7%** vs girls 2%; by 6 years, boys 2% and girls 7%.
- **Circumcision** lowers UTI in the first 6 months **~10-fold**. The number needed to circumcise is **111** normal boys, 11 with recurrent UTI and **4** with high-grade VUR.
- The **faecal reservoir** recycles the same strain, and antibiotics such as TMP-SMX destroy protective flora.
- **VUR** is found in 1–2% of newborns and **25–40%** after a first UTI. Most pyelonephritis occurs without VUR, but grade III–IV doubles pyelonephritic change.
- Treating **bladder and bowel dysfunction (BBD)** cuts recurrence and aids VUR resolution.
- In **neurogenic bladder**, 40–80% on CIC are bacteriuric; **ASB without VUR needs no antibiotics**, and frequent catheterisation prevents UTI.
- **Catheter-associated UTI** is the commonest nosocomial infection, so remove catheters early. Early infancy (IgG nadir) is the highest-risk period.

KEY POINT

Host factors either raise inoculum/dwell time (prepuce, faecal flora, stasis, constipation, catheter) or lower defence (immature immunity, receptor-bearing blood groups); prevention reverses whichever applies.

4 · Classification

- UTI is **complicated** (anomaly, dysfunction, foreign body; all neonatal UTI) or uncomplicated; upper or lower; first or recurrent.
- Febrile UTI is presumed pyelonephritis, but only **50–66%** of febrile children show acute changes on DMSA scan, the gold standard.

Recurrence	Between episodes	Cause	Remedy
Unresolved bacteriuria	Never sterile	Non-compliance, malabsorption, resistance	Culture-directed retreatment
Bacterial persistence	Sterile; same organism	Nidus: stone, anomaly, papilla, catheter, stent	**Remove nidus**
Reinfection	Sterile; new organism	New ascent	Prevention

Asymptomatic bacteriuria

- Kass definition: two cultures of **≥10^5 colony-forming units (CFU)/mL** of the same organism, without symptoms.
- Infants with ASB are treated and scanned. In school-age girls without VUR or anomaly, **do not screen or treat**, because this selects resistance.

5 · Diagnosis

Clinical

- Fever is ≥38 °C. A toxic look, age <30 days and ≥39.4 °C predict bacteraemia.
- Under 2 years, risk factors are fever >40 °C, previous UTI, suprapubic tenderness and an **uncircumcised penis**. **Another fever source, such as otitis, only halves UTI risk.**

Urine collection

Method	Contamination	Role
Bag	50–60% (false-positive culture up to 75%)	**Reliable only when negative**; never culture
Clean catch	~26%	Toilet-trained; preferred by NICE
Catheter	10–15%	Sensitivity 95%, specificity 99%; **standard <2 years non-toilet-trained**
Suprapubic aspiration (SPA)	<1%	Reference; US guidance gives 80–90% success

- Collect **before antibiotics** unless delay is dangerous. For 2–24 months the AAP accepts bag urinalysis; if positive, culture a catheter or SPA sample.

Test	Sensitivity	Specificity	Use
Leukocyte esterase (LE)	~79%	~87%	Negative helps **rule out**

Test	Sensitivity	Specificity	Use
Nitrite	~50%	**~98%**	Positive **rules in**
LE or nitrite or microscopy	99.8%	70%	Best rule-out
Pyuria ≥5 WBC/HPF	~74%	~86%	No advantage over LE
Gram stain, fresh urine	~91%	92–96%	Most reliable rapid test

- **Nitrite is falsely negative** with short bladder dwell (infants), dilute urine and gram-positives (Enterococcus).
- UTI without pyuria is rare, which separates it from ASB. A threshold of **≥10 WBC/HPF** works better under 2 years.
- **Culture is essential** unless urinalysis is entirely normal.

Guideline	Specimen	Significant count (CFU/mL)
Kass	Voided	$\geq 10^5$
AAP 2011 (2–24 months)	Catheter/SPA	**≥50,000 + pyuria**
EAU	Midstream/catheter	10^3–10^4 monoculture
RCH Melbourne	SPA / catheter	Any gram-negative / $>10^3$
ISPN (Italy)	Catheter / midstream	$>10^4$ / $>10^5$

- **The cleaner the specimen, the lower the accepted count.** Mixed growth and skin flora suggest contamination.
- CRP is sensitive, not specific. **Procalcitonin ≥0.5 ng/mL** best predicts acute pyelonephritis (APN) and later scarring.

6 · Imaging after UTI

- Routine imaging has not been shown to prevent scarring or renal failure, so **order a test only if it changes management**.
- US is abnormal in ~15% after a first febrile UTI, but only 1–2% need action. It is **urgent if the child is unwell or not responding at 48 h**.
- VCUG is the gold standard for VUR, done with sterile urine in a well child; CeVUS is an alternative.
- DMSA shows APN best within **1 week**; a defect at **≥6 months** is a scar, which cannot be told from dysplasia. CT is only for occult stones or mass vs inflammation.

Guideline	Ultrasound	VCUG	Late DMSA
AAP (2–24 months)	Yes, after first febrile UTI	If US abnormal (or recurrence)	No
NICE <6 months	Yes	Atypical/recurrent	Atypical/recurrent
NICE >6 months	Atypical/recurrent	No	Atypical/recurrent
Top-down	No	If acute DMSA abnormal	If acute DMSA abnormal
RCH Melbourne	If <3 months or seriously ill	No	No

- **NICE "atypical" UTI** means seriously ill, poor urine flow, abdominal or bladder mass, raised creatinine, septicaemia, no response within **48 h**, or a non-E. coli organism. Late DMSA is at 4–6 months.

- NICE and AAP miss about half of grade III–V VUR but spare most children a VCUG, because high-grade VUR declares itself at recurrence.

7 · Treatment

- **Delay matters**: new-scar odds double if antibiotics start after 72 h rather than within 48 h, so start empirically after urine collection.
- **Oral therapy equals intravenous** for non-toxic children over 2 months who tolerate oral drugs.
- **Admit** if under 1 month, toxic, dehydrated, vomiting or of doubtful compliance. Neonates need a septic work-up and IV therapy, switching to oral after ≥24 h afebrile.
- 90% are afebrile by 48 h. **If not improving at 48 h, do RBUS and broaden cover**.

Infection	Duration
Febrile UTI	**7–14 days** (shorter is inferior)
Afebrile cystitis	**2–4 days** (beats single dose; equals 7–14 days)
Lobar nephronia	≥3 weeks
Renal abscess	Antibiotics; drain if no response
Pyonephrosis	Antibiotics and drainage

- Choose by the **local antibiogram**. TMP-SMX and amoxicillin are poor empirical agents; prefer **cephalexin**, and nitrofurantoin for cystitis. **Proteus is a boys' organism**.
- **Neonates** need Enterococcus cover, so give **ampicillin with a third-generation cephalosporin or an aminoglycoside**.
- Use broad cover after previous UTI, recent antibiotics or admission, or with an anomaly. Fluoroquinolones are reserved for complicated UTI.

Agent	Dose (mg/kg/day)	Restriction
Cephalexin	50–100 in 4	—
Cefixime	8 once daily	—
Amoxicillin–clavulanate	20–40 in 3	—
TMP-SMX	6–12 (TMP) in 2	**<6 weeks/premature**: kernicterus
Nitrofurantoin	3–5 in 2	**<3 months**, GFR <50%, G6PD deficiency; **never for febrile UTI**
Ceftriaxone (IV)	75 once daily	—
Gentamicin (IV)	7.5 (once daily acceptable)	Nephro-/ototoxicity

No repeat culture is needed if the organism was sensitive.

8 · Renal scarring and sequelae

- Scars favour the **poles**, where compound papillae allow intrarenal reflux, and risk rises with VUR grade.

Pyelonephritic episode	1st	2nd	3rd	4th	5th
Scar risk	5%	10%	20%	40%	60%

- Childhood UTI causes **<1% of CKD**. Scars lead to hypertension and pre-eclampsia, so bilateral scars need lifelong checks of BP, function and proteinuria.

9 · Prevention and prophylaxis

- Recurrence is **10–30%**, highest in the first 3–6 months.
- **Continuous antibiotic prophylaxis (CAP)** is not given after a first UTI (AAP, NICE). It is for the **high-risk** child: girls with dilating VUR (≥III), BBD or recurrent febrile UTI.
- Give about **¼ of the therapeutic** dose at bedtime, using a different agent from treatment. Options are TMP-SMX (not under 6 weeks), nitrofurantoin and a cephalosporin in young infants.
- With VUR and BBD, recurrence is ~45% vs 15%; treat BBD with timed voiding and laxatives, then anticholinergics, biofeedback or CIC.
- **Circumcision** helps high-risk boys (recurrent febrile UTI, high-grade VUR, hydroureteronephrosis). Cranberry and probiotics are at most adjuncts.

10 · Special infections

P1 Xanthogranulomatous pyelonephritis. / Fungal UTI — XGP: Proteus, stone–obstruction triad, mimics tumour, nephrectomy (diffuse) vs partial (focal), hard dissection; Candida risks, C. glabrata resistance, treat >10,000–15,000 CFU/mL, fungus ball drainage + amphotericin

Not in the paper answer:

- In children, XGP has calculi in only 38–70% and presents with failure to thrive and anaemia.
- **Lobar nephronia** mimics a mass on CT and needs antibiotics for ≥3 weeks. **Pyonephrosis** must be **drained**.
- **Viral haemorrhagic cystitis** is due to adenovirus 11; **BK virus** after marrow transplant is treated with **cidofovir**.
- **Prepubertal epididymitis** is usually sterile; if culture-positive or recurrent, look for ectopic ureter or utricle. **Balanoposthitis** needs circumcision only if recurrent.

11 · Landmark trials

Trial	Design	Result	Take-home
RIVUR	607 children, VUR I–IV, TMP-SMX vs placebo, 2 years	Recurrent UTI ~halved (number needed to treat 8); **no difference in scarring**	Prevents recurrence, not scars; most benefit with BBD, febrile index UTI
Swedish Reflux Trial	Randomised, dilating VUR, 1–2 years	Prophylaxis reduced recurrence in girls	CAP for girls with dilating VUR

38 Core Principles of Perioperative Management in Children

1 · Perioperative surgical home and enhanced recovery

- **Enhanced recovery after surgery (ERAS)** combines counselling, shorter fasting, judicious IV fluid, no routine tubes or drains, opioid-sparing analgesia and early feeding and mobilisation. It shortens stay with no more complications.

2 · Preoperative assessment

- Risk is raised by upper respiratory tract infection (URTI), asthma exacerbation, dehydration, uncontrolled diabetes, prematurity and fever.
- **Preoperative anxiety** causes more pain and emergence delirium; explanation and play help.
- **Investigations are indication-driven**, with no routine bloods, chest X-ray or ECG. Check urine hCG in females of menstrual age; it can be falsely positive after augmentation cystoplasty.
- Send a **urine culture** before any procedure opening the urinary tract and reschedule until sterile. Treat asymptomatic bacteriuria before endourology.
- **URTI** leaves the airway hyperreactive for **6–8 weeks**, risking laryngospasm and bronchospasm. Postpone elective surgery at least 4, ideally 6 weeks.
- **Sepsis**: give antibiotics within **1 h** of septic shock and avoid neuraxial blocks. In **trauma**, assume a full stomach.

3 · Fasting (NPO, nil by mouth)

Minimum fast	Material
2 h	Clear liquids, carbohydrate drinks (European and Canadian 2022: 1 h)
4 h	Breast milk
6 h	Infant formula, non-human milk, light meal, tube feeds
8 h	Regular meal

- Non-human milk counts as a light meal. List the youngest first, because they dehydrate faster.
- **Diabetes**: list the child first, stop metformin 24 h before and halve the long-acting insulin the night before; a long case needs IV insulin.
- Omit ACE inhibitors that morning, stop warfarin and aspirin 7–10 days before, and never stop clonidine abruptly.

4 · Anaesthesia considerations

Neonate and preterm infant

- **Postoperative apnoea** peaks at 12–23 h, so give 23-h monitoring if postconceptional age is **< 50–60 weeks** (ex-preterm) or **< 44 weeks** (term). Defer elective surgery beyond these ages.

Anaesthetic neurotoxicity

- The FDA warning (2016) says repeated or lengthy (**> 3 h**) anaesthesia at **< 3 years** may affect brain development. Human trials of a single short exposure are negative (§11), so do not delay necessary surgery.

Comorbidities

- **Asthma**: give a β2-agonist before induction. **Chronic kidney disease**: avoid surgery within 4–6 h of haemodialysis and give no NSAIDs in CKD 3–4.
- **Sickle cell disease**: above all avoid dehydration, and also hypoxia, cold and tourniquets; acute chest syndrome is the main cause of death.
- **Phaeochromocytoma**: give α-blockade before β-blockade, with volume loading, and expect hypoglycaemia after resection.
- **Congenital adrenal hyperplasia**: give IV stress-dose hydrocortisone with glucose-containing fluid.

Spina bifida

- Confirm VP shunt function. **Latex allergy** can cause anaphylaxis, so care should be latex-free from birth.
- **Autonomic dysreflexia** (lesion at or above **T6**): bladder distension causes severe hypertension and bradycardia. Stop the stimulus, drain the bladder, sit the child up and give a rapid antihypertensive.

5 · Fluid and electrolyte management

Hypotonic fluids can cause fatal hyponatraemia.

- **Holliday–Segar 4-2-1 rule**: **4 mL/kg/h** for the first 10 kg, plus **2 mL/kg/h** for the next 10 kg, plus **1 mL/kg/h** per kg above 20. For 22 kg this is 62 mL/h.
- Add dextrose under 2–4 months; give no potassium if dialysed.

Maintenance fluid	Historical	Current
Tonicity	Hypotonic (0.45% saline)	Isotonic (0.9% saline or lactated Ringer) with dextrose
Potassium	10 mEq/L if < 10 kg; 20 mEq/L if > 10 kg	

KEY POINT

Why isotonic? Surgical stress, pain, nausea and hypoxia trigger non-osmotic ADH release, so hypotonic fluid adds free water and causes hyponatraemia (< 135 mmol/L). A child's large brain-to-skull ratio makes the resulting encephalopathy potentially fatal.

6 · Blood loss and transfusion

- Estimated blood volume (EBV) is 80–90 mL/kg in a term neonate and 70–75 mL/kg in a child. Maximum allowable blood loss = EBV × (starting − target haematocrit) / starting haematocrit.
- Transfuse at haemoglobin **< 7 g/dL** in stable children; 10–15 mL/kg raises haemoglobin by about 2–3 g/dL.

7 · Temperature

- Infants cool quickly, causing acidosis, coagulopathy and apnoea, so use a warm theatre, forced-air warming and warmed fluids.

8 · Antibiotic prophylaxis and intraoperative crises

- **Timing**: give within **60 min** of incision; vancomycin and fluoroquinolones within **120 min**.
- Doses: cefazolin **30 mg/kg** (redose at 4 h); ampicillin 50 mg/kg (redose at 2 h); gentamicin 2.5 mg/kg, vancomycin 15 mg/kg, metronidazole 15 mg/kg and ciprofloxacin 10 mg/kg (not redosed).

	Anaphylaxis	Malignant hyperthermia (MH)
Triggers	Neuromuscular blockers, β-lactams, latex	Succinylcholine, volatile agents (not nitrous oxide)
First sign	Collapse, bronchospasm, hives minutes after induction	Unexplained hypercarbia, tachycardia, masseter spasm; late — acidosis, hyperkalaemia, rigidity
Treatment	**Adrenaline**, 100% oxygen, fluids; send tryptase	Stop triggers, **dantrolene**, clean circuit; admit 24–48 h (recrudescence)

9 · Analgesia

- Under 3 years, use the FLACC pain score.
- **Regional blocks under general anaesthesia** are safe and opioid-sparing (penile, **caudal**, epidural). Caudal 0.25% bupivacaine is 0.5–1 mL/kg; maximum bupivacaine is 2.5 mg/kg.
- Give **multimodal, scheduled** analgesia: acetaminophen 15 mg/kg 6-hourly and ibuprofen 10 mg/kg 8-hourly. Ketorolac is limited to 72–96 h, and no NSAIDs are given under 6 months.
- **Codeine is contraindicated**, because ultrarapid metabolisers make lethal amounts of morphine. Limit discharge opioids to 3–5 days.

10 · Postoperative care, VTE and day-case discharge

- **Emergence delirium** occurs at 2–8 years, within 30 min of waking; the child does not recognise parents, unlike a child in pain.
- **Venous thromboembolism** is rare before puberty; at-risk adolescents need mobilisation and mechanical prophylaxis, with LMWH if several factors are present.
- **Day-case** surgery suits ASA I–II children outside the apnoea-risk ages having a short, low-blood-loss procedure.

11 · Landmark trials

Trial	Design	Result	Take-home
GAS	Randomised; infant hernia, awake regional vs sevoflurane < 1 h	No cognitive difference at 2 and 5 years	Single short anaesthetic safe
PANDA	Sibling-matched; single exposure < 3 years	No IQ difference at 8–15 years	Repeated exposure untested
MASK	Cohort; exposure < 3 years	Negligible deficit after single exposure	Reassuring

39 Principles of Laparoscopic and Robotic Surgery in Children

1 · Evolution and indications

P3 Discuss diagnostics and operative pediatric laparoscopy in urology — evolution 1976, 1994, 1995, 2002; indications; age IAP limits; only contraindication; access; complications (right common iliac, inferior epigastric, port-site hernia 0.5–3%)

Not in the paper answer:

- The first paediatric urological laparoscopy was for non-palpable testis in **1976**. The wristed robot removed the suturing bottleneck, so **reconstruction** grew, with pyeloplasty the commonest.
- The only strict contraindication is **inability to tolerate pneumoperitoneum**; age and size are not contraindications.

2 · Physiology of pneumoperitoneum

P1 Pneumoperitoneum physiology; laparoscopic access and gases — pressure vs CO2 effects on heart, lungs, kidney, ICP;
vagal bradycardia; CO2 choice;
gas embolism signs and treatment (left lateral head-down, oxygen, aspirate)

Not in the paper answer:

- Effects depend on pressure and reverse on desufflation, so use the **lowest age-appropriate pressure**: intra-abdominal pressure (IAP) < 12 mm Hg, and about 7 mm Hg is feasible.
- Infant oliguria is transient and is not a guide to resuscitation, so give **no fluid boluses**.

Age	0–1 month	2–12 months	1–2 years	2–8 years
Maximum IAP	**≤ 6**	**≤ 8**	**≤ 10**	**≤ 12 mm Hg**

- **Cardiovascular**: IAP > 10 mm Hg lowers venous return and cardiac index. Insufflation can trigger a **vagal reflex**, so desufflate and reinsufflate slowly.
- **Respiratory**: infants have a compliant chest wall and low FRC, so tidal volume falls, worse with Trendelenburg, and CO2 absorption affects them most.
- Laparoscopy is generally safe with **VP shunts**. Infants cool fast, so use warmed humidified gas.

3 · Access

- Open (Hasson) and closed (Veress) entry do not differ in injury rates (Cochrane); confirming intraperitoneal position is what protects.
- After scarring or failed umbilical entry, use **Palmer's point** (left midclavicular line, about 3 fingerbreadths below the costal margin).

4 · Ports, instruments and placement

- **3- and 5-mm** laparoscopic ports suit infants, whereas robots need **8 mm**. Robotic wristed instruments have 7 degrees of freedom against 4 for laparoscopic ones.
- Triangulation is essential in laparoscopy, while the robot tolerates linear placement. **HIdES** puts ports below the Pfannenstiel line.

5 · Approaches

	Transperitoneal	Retroperitoneal	Vesicoscopic
Space	Largest; familiar	Limited	Limited; small bladder predicts complications
Strength	Complex reconstruction	Avoids peritoneum; preserves peritoneal dialysis	Intravesical access
Limitation	Colon mobilised or mesenteric window	Unfamiliar view; peritoneal tear collapses space	Learning curve; port-site closure
Uses	Testis, kidney, ureter, bladder	Pyeloplasty, (hemi)nephrectomy, lymph node dissection	Cross-trigonal reimplantation

- In **left pyeloplasty**, a transmesenteric window avoids colonic mobilisation.

6 · Robotics in children

- **Working space** is about **1 L at 1 year vs 6 L in adults**, so arms collide; wide port spacing and bowel preparation help.
- **Limitations**: there are no 5-mm robotic ports, and single-port needs a 25-mm cannula and about **10 cm** to the target, which is prohibitive in small children.
- **Benefits** are shorter stay and less narcotic use; incisions under 15 mm contract with growth.

7 · Complications

- The major vascular injury is to the **right common iliac** artery at Veress entry; the minor one is the **inferior epigastric**, so insert ports under vision.
- Port-site hernia peaks around 1 year, so **close fascia** at every port.
- Experience is the strongest predictor, with fewer complications above 12 cases a year.

> **KEY POINT**
> **Ergonomics**: set the table at 70–80% of elbow height and the monitor at eye level. The robot offloads the arm but fixes the neck, so use the clutch to keep the elbows on the armrest.

40 Clinical and Urodynamic Evaluation of Lower Urinary Tract Dysfunction in Children

1 · Principles and terminology

- Non-invasive assessment excludes anatomical or neurological causes, then defines **storage vs emptying** dysfunction (ICCS 2014).
- **Expected bladder capacity (EBC)**: infant ≤1 y = **weight (kg) × 7 mL**; older child = **(age in years + 2) × 30 mL**. Small is <65% EBC; large is >150% (some use >130%).

2 · History

- **Secondary enuresis** is wetting after **≥6 months** dry, so look for stressors.
- In girls, poor **toilet posture** activates the pelvic floor and causes incomplete emptying; a footstool helps.
- **Holding manoeuvres** (potty dance, squatting) suppress detrusor overactivity (DO).
- Bowel dysfunction may be the sole cause of urinary symptoms. Target a painless **Bristol 3–4 stool daily**.
- **Rome IV functional constipation** (developmental age ≥4 y, ≥1 month) needs **≥2 of**: ≤2 stools/week; ≥1 soiling/week; retentive posturing; painful or hard stools; large rectal mass; stools blocking the toilet.
- **Neurological red flags** are perineal numbness, or gait or limb change with new LUTS.

ALGORITHM

1. **Bowel and bladder dysfunction (BBD)**
2. Apply Rome IV; obtain a 7-day diary, urinalysis and pelvic US.
3. **Treat the bowel first.**
4. If LUTS persist, do uroflow with EMG and start urotherapy.
5. If that fails, add an anticholinergic, then a β3-agonist; then botulinum toxin, biofeedback or neuromodulation.

- The **48-h frequency–volume chart** shows capacity and polyuria and is **mandatory** when first-line therapy fails.
- Scores: **DVSS** best tracks treatment, and **DVISS** is best for diagnosis.

3 · Physical examination

- Measure height, weight and BP at **every visit**.
- **Neurological**: include perineal sensation, anal tone and sacral reflexes (anal wink, bulbocavernosus); their absence means the S2–S4 arc is interrupted.
- **Spine**: a simple sacral dimple (<2.5 cm from anus) is low risk and gets spinal US up to 6 months. High-risk signs (hair tuft, lipoma, vascular malformation, multiple stigmata, VACTERL) need **MRI**.
- **Genitalia**: **continuous incontinence in a girl suggests an ectopic ureteric orifice.**

4 · Laboratory tests

- Do a **urinalysis for every child with LUTS**; **culture is not routine.** On CIC, culture only with fever, pain, new incontinence or spasm.

5 · Non-invasive urodynamics

P3 P1 Uroflowmetry; EMG — bell/tower/staccato/interrupted/plateau patterns and meaning; Qmax peak ≥2 s; patch EMG at 3 and 9 o'clock; lag time <2 s DO, 2–6 s normal, >6 s bladder-neck dysfunction

Not in the paper answer:

- Uroflow is valid only if toilet-trained and voiding **≥50% EBC** (replacing the adult ≥150 mL rule), with more than one curve.
- A staccato curve never reaches zero, whereas an interrupted one does.

> **KEY POINT**
> The curve reflects detrusor against outlet. Too much detrusor gives a tower; too much outlet gives a plateau; intermittent outlet activity gives staccato; intermittent detrusor effort gives an interrupted curve. Patch EMG tells which side is at fault.

- **Pelvic US**: the bladder wall is abnormal if **>3 mm full / >5 mm empty**. A rectal diameter **>30 mm** without urge means faecal loading.
- **PVR** is abnormal at 4–6 y if **>30 mL or >21%** of capacity and at 7–12 y if **>20 mL or >15%**. It is valid only **<5 min** after voiding with the bladder at 50–115% EBC, so repeat it.

6 · Invasive urodynamics

Filling pressure, not wetting, decides renal risk.

Indications

- **Neurogenic**: spinal dysraphism, tethered cord, sacral agenesis, transverse myelitis or spinal cord injury (SCI).
- **Severe obstruction**: posterior urethral valves (PUV) or urethral atresia.
- Non-neurogenic LUTS: study only after failed 2nd/3rd-line therapy, and only if it will change management.

Technique

- **Avoid sedation or general anaesthesia**; if unavoidable, use ketamine rather than midazolam.
- Routine prophylaxis is not needed, because post-study UTI is about **1.4%**.
- Use a **6 Fr dual-lumen** urethral and a rectal catheter, and **fill at 5–10% of EBC per minute** with warm fluid.
- **Stop filling** at strong urge, discomfort, voiding, leak greater than fill rate, Pdet persistently **>40 cmH_2O**, or volume **>150% EBC**.
- Report the first fill, which gives the highest Pdet. **Videourodynamics** adds trabeculation, VUR with its onset volume and pressure, and bladder-neck shape.

Storage phase

- **Pdet = Pves − Pabd**, and normal filling is flat. **DO** means phasic involuntary rises.
- **Compliance = ΔV/ΔPdet**: **>20 normal; 10–20 moderately; <10 severely impaired**. In small contracted bladders absolute Pdet (<30–40) is the better guide.

P1 Leak point pressures (VLPP, ALPP, DLPP) in SUI: clinical implications — DLPP = outlet resistance, high = upper-tract risk; ALPP = sphincter competence, low = bad; McGuire 1981

Not in the paper answer:

- A DLPP **>40 cmH₂O**, or no leak at high pressure, signals danger; the adult ALPP cut-off (<60 cmH₂O) has no paediatric norms.

> **KEY POINT**
> **McGuire (1981)**, myelodysplasia: with leak pressure >40 cmH₂O, hydronephrosis or secondary VUR developed in **85%**. Aim for the lowest achievable pressure.

Voiding phase

- **High Pdet with low Qmax means obstruction.**
- Anatomical obstruction has a **quiet EMG**; functional obstruction (DSD) has an **active EMG**.
- Normal Pdet at Qmax is about 66 cmH₂O in boys and 57 in girls; infant boys reach 118 physiologically MCQ.

> **KEY POINT**
> **Upper-tract threats**: compliance <20 (especially <10); DLPP >40 cmH₂O; sustained high filling Pdet; DSD with poor emptying; high voiding pressure with poor flow.

- **UMPIRE** (spina bifida) studies at about **3 months**, then at 1, 2 and 3 years, then yearly to 5 y if VUR or a hostile bladder. Hydronephrosis is a late sign, so re-study on any change and before reconstruction.

7 · Evaluation in specific conditions

Condition	Typical findings	When to study
Non-neurogenic LUT dysfunction / enuresis	DO, dysfunctional voiding	Only if refractory
Spinal dysraphism	UMPIRE risk by DLPP/end-fill Pdet: **<25 low; 25–39 or DO intermediate; ≥40 high**	Baseline, serial, pre-surgery
Sacral agenesis	Areflexia with weak sphincter, or DO with DSD; level does not predict	If voiding dysfunction
Anorectal malformation	Tethered cord common	Baseline about 1 y after repair
PUV	DO, then poor compliance, then myogenic failure ("valve bladder" about 15%)	Serial
Cerebral palsy	Upper motor neuron; DO, DSD rare	Only if conservative therapy fails
SCI	Spinal shock causes retention; higher lesions have more DSD	Video-urodynamics 3–6 months after injury
Transverse myelitis	DO most often; motor recovery does not predict bladder	Early, with CIC

> **KEY POINT**
> UMPIRE dropped DSD from high risk because reviewers disagreed on who had it, not because DSD is benign. Neurogenic bladder goals are normal capacity, low storage pressure, complete emptying and upper-tract protection.

8 · Landmark trials

Trial	Design	Result	Take-home
McGuire 1981	Myelodysplasia urodynamic cohort	DLPP >40 cmH_2O led to upper-tract damage in 85%	40 cmH_2O = safety threshold

42 Bladder Anomalies in Children

1 · Classification and embryology

P3 Embryological development of urinary bladder and management of Hutch Diverticulum? — cloacal division weeks 4–6, UGS fates, trigone, urachus months 4–5; anomaly table; Hutch primary vs secondary, VCUG, treat outlet first, diverticulectomy + reimplant

Not in the paper answer:

- Anomalies are **structural** (exstrophy, duplication, agenesis, urachal, diverticula) or **functional**; prune-belly syndrome (PBS) is both.

2 · Fetal bladder on ultrasound

- The fetal bladder empties every **15–20 min**, so if unseen, **repeat in the same sitting**. Amniotic fluid is mainly fetal urine after **16 weeks**.
- In exstrophy there is no filling with **normal amniotic fluid**.

	Agenesis	Hypoplasia	Exstrophy
US	Absent after repeat scan	Small, never reaches capacity	No filling; bladder template on abdominal wall
Core defect	Cranial UGS atrophy or no trigone	Inadequate filling or storage	Open bladder plate
Outcome	Survives only with ectopic ureteric drainage	May grow; augmentation often	Reconstruction (see Ch 43)

- **Agenesis** is very rare (about 1:600,000, mostly female); ureters drain into müllerian structures or the rectum, and it is treated by ureterosigmoidostomy or stoma.

3 · Megacystis

- In classic congenital megacystis, **massive VUR causes the dilation**; today the term means any prenatally dilated bladder.

Gestation	Normal	Abnormal
10–14 weeks	Length ≤6 mm; <10% of CRL	**≥8 mm**; ≥13% of CRL
15–40 weeks	Length about GA − 5 mm	> GA + 12 mm
Later (functional)	Empties within 45 min	No emptying in 45 min

- **Causes**: LUTO, with PUV in about 50%, urethral atresia or anterior valves; non-obstructive causes are PBS, MMIHS and cloacal anomalies. Male:female is 8:1.
- **Aneuploidy is about 1/5** (trisomy 13/18), so karyotype.
- In euploid fetuses, 7–15 mm mostly resolves, while **>15 mm** never does. **Persistent megacystis with oligohydramnios means LUTO.**

MMIHS

- MMIHS is a visceral myopathy (**ACTG2**), mostly female, with megacystis and normal or increased fluid. The bladder is acontractile and most need **lifelong CIC**; TPN complications are the leading cause of death.

KEY POINT

A female fetus with megacystis cannot have PUV, so think of MMIHS, cloacal anomaly or urethral atresia.

4 · Bladder duplication

- **Complete sagittal duplication is commonest**, with external genital duplication in up to 90% and lower GI duplication in up to 42%.

ALGORITHM

1. Early: relieve obstruction and prevent infection.
2. Later, if both sphincters are competent, then join the distal urethras.
3. If one is incompetent, then close that bladder neck and excise its urethra.
4. Join duplicated vaginas, with vulvoplasty.

- **Sphincter competence, not anatomy, decides reconstruction.**

5 · Bladder diverticula

- A primary (Hutch) diverticulum is observed if small and excised if there is VUR. For a secondary one, **treat the outlet first**.
- **Syndromic** diverticula (**Ehlers–Danlos, Williams, Menkes**) are multiple and giant; resect only for UTI or stones, as recurrence is common.

6 · Urachal anomalies

	Patent urachus	Sinus	Cyst	Vesicourachal diverticulum
Open	Whole canal	Umbilical end	Isolated segment	Bladder apex
Presents	Urine from the umbilicus in a neonate	Draining umbilicus	Infection (S. aureus), painful mass	Incidental
Imaging	US; fistulogram or VCUG	Sinugram	US; CT/MRI if extensive	Incidental
Treat	Excise with cuff	Complete excision	Drain, then excise	Only if a narrow neck causes stones or UTI

- The differential is a patent **omphalomesenteric duct**. Incidental remnants are increasingly excised only if symptomatic.

ALGORITHM

1. If infected, drain and give antibiotics first.
2. Infraumbilical incision; circumscribe the tract, preserving the umbilicus.
3. Dissect extraperitoneally and excise with a **bladder cuff**.
4. Two-layer closure; catheter overnight.

7 · Acquired lesions

Lesion	Key features	Treatment
Haemangioma	Klippel–Trénaunay, Sturge–Weber; haematuria	Fulguration or Nd:YAG laser

Lesion	Key features	Treatment
Fibroepithelial polyp	2nd commonest paediatric bladder tumour	Endoscopic resection
Nephrogenic adenoma	Female 5:1; after infection, surgery, augmentation	TUR; **80% recur**, so surveillance cystoscopy
Cystitis cystica	Almost all female; mimics tumour	Long-term antibiotic prophylaxis
Eosinophilic cystitis	Male	**Biopsy mandatory**; steroids and antihistamines
Haemorrhagic cystitis	Cyclophosphamide or ifosfamide; BK virus, adenovirus	Mesna prevents; pentosan polysulfate

43 Exstrophy–Epispadias Complex

1 · Spectrum, epidemiology and embryology

Rupture of the cloacal membrane explains the whole spectrum.

P3 P1 Bladder and ureter development, exstrophy–epispadias complex and its management; Describe embryology of bladder… theories for EEC… skeletal abnormalities — spectrum, CBE 1 in 46,000, M:F 2.3:1, offspring 1 in 70; 22q11.2, ISL1, p63; IVF/progesterone/smoking/folate; failed mesodermal reinforcement of cloacal membrane, Marshall–Muecke and other theories; pelvis 12°/18° rotation, anterior pubis 30% short, diastasis 4.8 cm; MSRE stages, osteotomy indications, CPRE outline

Not in the paper answer:

- Spica cast fails in 66% without osteotomy, and diastasis >6 cm needs staged closure.
- Rupture timing explains the phenotypes: early, extensive rupture opens bladder **and** hindgut (cloacal exstrophy); later rupture opens the bladder only; rupture at the genital tubercle gives epispadias.

2 · Anatomy of classic bladder exstrophy

- **Pelvis**: the acetabula are retroverted and the posterior segment is normal. The resulting waddling gait and out-toeing improve with growth.
- **Pelvic floor**: the levator ani lies posterior and flattened. Continent patients have the shortest intersymphyseal distance and deepest bladder neck, hence **deep placement, osteotomy and pelvic fixation**.
- The **intrasymphyseal band** tethers the posterior vesicourethral unit to the pubic rami. It contains **no striated sphincter**, so radical incision down to the levator hiatus is essential.
- **Indirect inguinal hernia** is common, often bilateral in males, so repair it at closure.
- **Anorectum**: the anus is anterior. **Rectal prolapse before closure** resolves with closure, but prolapse **after closure** suggests outlet obstruction and needs urgent cystoscopy.
- **Male**: the **anterior corpora are about 50% shorter** and 30% wider. The prostate is never circumferential. Cavernous nerves lie laterally, so potency is retained, though retrograde ejaculation may follow BNR.
- **Female**: the vagina is short with a stenotic anterior orifice, the **cervix sits in the anterior vaginal wall** and the clitoris is bifid. A wide levator hiatus causes **uterine prolapse**, even in nulliparous women, and **osteotomy does not prevent it**.
- **Bladder**: it is **immature, not neuropathic**. The exposed plate develops polyps, squamous metaplasia and cystitis glandularis, which carry an **adenocarcinoma risk**. Compliance falls after BNR, so follow-up is lifelong.
- The ureters enter laterally without obliquity, so **vesicoureteral reflux (VUR) occurs in 100%** of closed bladders and is corrected by reimplantation at BNR.

3 · Variants

Except superior vesical fissure, variants are managed like CBE.

Variant	Feature	Management
Skin-covered (split symphysis), the commonest	Musculoskeletal defect; urinary tract intact	Formal closure at birth

Variant	Feature	Management
Duplicate exstrophy	Everted mucosal patch plus a closed pelvic bladder	Excise patch, close wall
Superior vesical fissure	Rupture only at the top; sphincter intact	Toilet-trains normally; no BNR
Pseudoexstrophy	Musculoskeletal defect only; low umbilicus	—
Epispadias with bladder prolapse	Prolapse may show only on Valsalva	Repair, with or without osteotomy

4 · Prenatal diagnosis

- Only about 25% are diagnosed prenatally. Signs are **absent bladder filling**, a low umbilicus, wide pubic rami, small genitalia and an **enlarging lower abdominal mass**. Plan **delivery at an exstrophy centre**.

5 · Newborn care and selection

A good first closure best predicts continence.

ALGORITHM

1. Tie the cord with 2-0 silk (no clamp, which traumatises mucosa).
2. Cover the plate with non-adherent plastic wrap; irrigate with saline at each nappy change.
3. Assess heart and lungs; obtain a renal US.
4. Transfer to an exstrophy centre for MDT counselling.

- **Assess** the template (≥**3 × 3 cm** is suitable), diastasis, pelvic malleability under anaesthesia and penile length. A plate that **bulges on crying** is adequate.
- A plate <3 cm, polyp-covered or fibrotic is **inadequate**. Penoscrotal duplication, hypoplastic bladder and bilateral hydronephrosis preclude primary closure.
- A small plate is closed **later, at 6–9 months, with osteotomy**, because a bad closure ruins continence. If it has not grown by 6–12 months, excise it and divert. After delayed closure a **catheterisable channel is needed far more often**.

6 · Surgical philosophies

Repair	Stages / feature	Weakness
Modern staged repair (MSRE)	Closure at birth; epispadias repair at 6–12 months; BNR with reimplant at 4–7 years	Multiple operations
Complete primary repair (CPRE, Mitchell)	Newborn closure with penile disassembly and neck tailoring	Hypospadias; glans loss; early reimplant
Kelly	Closure, then radical soft-tissue mobilisation at 3–6 months, then hypospadias repair; avoids osteotomy	Fistula, stenosis
Erlangen	Total repair at about 8 weeks without osteotomy	—
Mainz	Closure, then sigma pouch at about 2 years	Colonic malignancy

KEY POINT

Erlangen, Kelly and CPRE claim to avoid BNR, yet almost all still need an outlet procedure. Compressing the reconstruction adds ischaemic and outlet risk without removing that need.

7 · Primary closure (MSRE stage 1)

Closure creates an incontinent epispadias with enough outlet for growth.

ALGORITHM

1. General anaesthesia with **epidural** (catheter to T10–T12).
2. Outline a 2-cm mucosal strip from the trigone to mid-shaft (boy) or vaginal orifice (girl); incise the bladder–skin junction.
3. Free the bladder from the rectus and strip peritoneum off the dome.
4. **Divide the urogenital diaphragm fibres sharply down to the levator hiatus**; free the corpora from the rami.
5. Close bladder, neck and posterior urethra; **taper over a 12-Fr sound**.
6. Place a suprapubic Malecot (4 weeks) and ureteric stents; leave the **urethra unstented**.
7. Close the pubis with a **No. 2 nylon horizontal mattress** suture, knot away from the neourethra; fashion a neoumbilicus.

- **Tapering** gives enough resistance for bladder growth and against prolapse, but not enough to harm the kidneys.
- **Ureteric stents** stay 10–14 days because oedema can obstruct the ureters; a urethral stent causes necrosis.
- The **commonest error** is inadequate urogenital diaphragm dissection, after which the unit is pulled forward and prolapses.

8 · Osteotomy and immobilisation

Osteotomy and rigid immobilisation cut closure failure several-fold.

- In a **newborn <72 h** whose malleable pubes meet easily, osteotomy is not needed. **If in doubt, osteotomise.**
- Osteotomise if closure is **>72 h** after birth, diastasis is **>4 cm**, the pelvis is stiff, the template is large, after **failed closure**, and in **cloacal exstrophy**. Diastasis >6 cm needs staged closure with a fixator.

Type	Position	Features
Combined bilateral anterior innominate and vertical iliac (preferred)	Supine	Best malleability; least recurrence
Bilateral posterior iliac (original)	Prone	Worse approximation, more recurrence
Anterior innominate alone	Supine	Less mobility

- Aftercare is **modified Buck traction for about 4 weeks with an external fixator for 4–6 weeks**. Without osteotomy, use **modified Bryant traction** (hips 90°) for 4 weeks.

Immobilisation (CBE)	Failure overall	With osteotomy	Without osteotomy
Spica cast	47%	30%	**66%**

Immobilisation (CBE)	Failure overall	With osteotomy	Without osteotomy
External fixation with traction	**7.5%**	8%	— (few)

- Failure falls with traction plus fixator for **>4 weeks**; osteotomy type and epidural make no difference. Complications are pin-site inflammation, transient femoral nerve palsy and delayed union.

9 · After primary closure

- At 4 weeks calibrate the outlet, scan and check the residual. Give **continuous antibiotic prophylaxis**, because all have VUR.
- Perform **yearly cystoscopy and cystogram under anaesthesia** to measure capacity. If capacity is **<30 mL at 1–2 years**, a continence procedure may not be feasible.
- If the **outlet is too tight**, escalate from dilation to incision and revision. Recurrent UTI with a distended bladder suggests **intrapubic stitch erosion**.

10 · Epispadias repair (stage 2)

Urethroplasty raises outlet resistance, so the bladder grows.

- It is done at 6–12 months, before BNR, because urethroplasty **increases capacity**. Give **testosterone enanthate 2 mg/kg intramuscularly (IM) at 5 and 2 weeks** before every urethroplasty.
- **Modified Cantwell–Ransley** (preferred): the plate is freed but its **distal 1 cm stays on the glans**; it is tubularised over 8 Fr and the **corpora are rotated over the neourethra**, which then lies ventral.
- **Chordee** is corrected by dorsomedial corporotomy with cavernocavernostomy; skin cover uses ventral preputial flaps.
- Early fistula occurs in about 1 in 6, usually at the base; **coronal fistula is rare**.
- **Mitchell–Bägli disassembly** risks plate ischaemia, and most end **hypospadiac**. Glans or corporal loss is linked to closure **without osteotomy**, so if ischaemia is seen, osteotomise immediately.
- **CPRE**: about half need reimplant in year 1, 65–80% need an outlet procedure, and most end on CIC.
- **Kelly**: unaided voided continence is under 1 in 5, with glans loss reported.
- **Combined closure and epispadias repair** suits failed or delayed closures and boys >6 months, but risks ischaemic penile loss.
- **Female closure**: bladder, urethra and vagina move deep as one unit, with **monsplasty and joining of the clitoral halves**.

11 · Bladder neck reconstruction (stage 3)

Capacity at BNR best predicts dryness.

- **Prerequisites**: capacity **≥100 mL** under anaesthesia, stable urodynamics, a motivated child of 5–7 years and a voiding programme. Reimplant earlier for pyelonephritis or worsening hydronephrosis.

ALGORITHM

1. **Modified YDL**
2. Open the bladder neck transversely.
3. Perform **Cohen cross-trigonal reimplantation**.
4. Outline a posterior mucosal strip **15 × 30 mm**; denude the lateral muscle triangles.
5. Tubularise the strip and **double-breast the muscle flaps** over it.
6. Suspend to the rectus fascia; suprapubic tube for 3 weeks, then clamp trials.

- After MSRE, about **70–75%** are dry by day (intervals >3 h); CPRE with YDL is lowest (about 56%).
- **Predictors** are template size, **successful primary closure** and, most importantly, **capacity at BNR**. Closure within 72 h or with osteotomy makes continence likely, whereas delayed closure without osteotomy gives about 10%. After two closures, fewer than 1 in 5 void continently.
- Day dryness comes within about 12 months and night dryness takes 2–3 years; **not dry by 2 years means failure**.

Category (after BNR)	Definition
Continent	Dry day and night, interval ≥3 h
Socially continent	Dry 3 h by day, occasional wet nights
Wet	Dry <3 h by day, wet at night

12 · Failures and complications

Each failed closure shrinks the bladder.

- **Failed closure** means dehiscence, **bladder prolapse**, outlet stricture or vesicocutaneous fistula. Causes are incomplete pelvic-diaphragm mobilisation, inadequate immobilisation, infection and tube malfunction.
- In expert series, major dehiscence or prolapse occurs in about 3% and femoral palsy in about 2%.
- **Reclosure** waits at least 4–6 months and uses **osteotomy**, rigid immobilisation and radical urogenital diaphragm division. After secondary closure, voided continence is only about 1 in 6.
- Outlet stricture presents with UTI, an enlarging bladder, stones or rectal prolapse; obstruction for more than 6 months usually means augmentation.
- **Failed BNR** means no 3-h dry interval by 2 years. **Redo YDL** only if the neck is patulous, capacity >100 mL and urodynamics stable; otherwise do **bladder neck transection, augmentation and a Mitrofanoff channel**.
- **Genital**: urethrocutaneous fistula is commonest; glans or corporal loss may need **radial forearm phalloplasty**.

13 · Diversion and augmentation

- **Ureterosigmoidostomy** needs anal continence. It risks pyelonephritis, hyperchloraemic acidosis and **colonic adenocarcinoma about three decades later**, so **yearly colonoscopy** is required.

- If the bladder is too small, use a **non-refluxing colon conduit**. After failed BNR, do **augmentation with an appendiceal channel, with or without neck closure**; stones are the commonest complication.
- **Continent diversion** is indicated for closures that do not grow, inadequate capacity or wetness after BNR, and upper-tract decompensation.

14 · Long-term issues

- **Malignancy**: augmented exstrophy bladders develop adenocarcinoma, so survey endoscopically from 5 years after reconstruction.
- **Male**: libido, erection and orgasm are preserved. Fertility is impaired by retrograde ejaculation and oligospermia; about 1/4 achieve paternity, half via ICSI.
- **Female**: a stenotic introitus needs **perineal flap vaginoplasty**, because a cutback worsens prolapse. **Uterine prolapse** is treated by sacral suspension. **Pregnancy** ends in a **planned caesarean with a urologist present**.
- Psychosocial problems need lifelong support and transition care.

15 · Epispadias

P3 Development of the urethra, congenital urethral anomalies and management of epispadias — incidence, male/Davis classification, VUR rates, >3 cm means exstrophy pathway, Cantwell–Ransley then YDL at ~100 mL, female repair, continence males 47–82% / females 70–87%

Not in the paper answer:

- Glanular epispadias is never incontinent.

	Male	Female
Incidence	1 in 117,000	1 in 484,000
Classification	Glanular, penile, **penopubic** (about 70%, incontinent)	Davis: vesicular, subsymphyseal, retrosymphyseal (incontinent), in rising severity
VUR	30–40%	30–75%
Presentation	At birth	May present only with incontinence

ALGORITHM

1. **Female repair (6–12 months)**
2. Incise vertically through the urethral wall at 9 and 3 o'clock.
3. Excise the urethral roof (dorsal wedge taper) and close over a 10-Fr Foley.
4. Unite the clitoral halves and labia minora; bring mons fat over the suture line.
5. Catheter 5–7 days; BNR at 4–5 years.

- Treat as exstrophy (osteotomy and closure) if diastasis is >3 cm, the sphincter is scant or capacity is small.
- In **males** with a small bladder, do urethroplasty first, then BNR at about 100 mL (4–5 years).
- Female continence after BNR is about **87%**, as good as in CBE girls, because the bladder was never exposed.

16 · Cloacal exstrophy

A multisystem defect; neurological level sets disability.

- Incidence is 1 in 200,000–400,000. **OEIS** stands for Omphalocele, Exstrophy, Imperforate anus and Spinal defects. **Two hemibladders flank exstrophied ileocaecal bowel**, with a rudimentary hindgut and separated hemiphalluses.

Feature	CBE	Cloacal
Spinal defects	Uncommon	**85–100%** (lipomeningocele commonest)
Omphalocele	Rare	Near-universal
Upper-tract anomalies	Uncommon	41–60%; solitary kidney commonest
Osteotomy	Selective	Always
Continence	Mostly voiding	Mostly augmentation with CIC

- Only about 1 in 5 has a sacral skin sign, so all need spinal MRI. A 46,XY child may have an **intravesical phallus**, so biopsy before calling it aphallia.
- Prenatally, the **"elephant trunk" sign** is prolapsed ileum. **Gender** follows **karyotype**.
- **Bowel**: never discard hindgut; tubularise the caecal plate with an end colostomy.

ALGORITHM

1. Stage 1 (newborn): close the omphalocele, separate the caecal plate, join the hemibladders and form a colostomy.
2. Stage 2: bladder closure with **combined osteotomy** and fixation for 6–8 weeks.
3. At 5–7 years: YDL only if capacity >100 mL (rare); usually augmentation with a continent stoma.

- Combined osteotomy gives about 90% closure success but no continence gain; staged closure beats single-stage.
- Primary closure fails in about 31–43%. Continence relies on augmentation with CIC, and the **neurological level** determines disability.

44 Prune-Belly Syndrome

P1 P3 Prune Belly Syndrome. Antenatal diagnosis, pathophysiology and postnatal management; Prune belly syndrome — triad, 1 in 29,000–44,000, 95% male; four theories, MYOCD/FLNA; pathology by organ (dysplasia ~50%, VUR ~75%, failed pump); extra-GU rates; Woodard I–III; neonatal work-up; vesicostomy, reconstruction after 3–6 months; orchidopexy ~6 months; Randolph/Ehrlich/Monfort; creatinine <0.7 mg/dL, 30% CRF

Not in the paper answer:

- Renal failure is driven by pyelonephritis, dysplasia and polyuria; transplant outcomes are equal.
- Female PBS is ~5%, with normal gonads.

1 · Definition and aetiology

- The **triad** (Eagle–Barrett) is deficient abdominal wall muscle, bilateral intra-abdominal testes and a dilated urinary tract. Incidence is ~**1 in 29,000–44,000** and **95% male**; it is commoner in twins and African ancestry.
- **Survival follows the kidneys and lungs, not the belly**; the wall is least prognostic.
- There is no inheritance pattern; multiplex families show the smooth-muscle genes **MYOCD** and **FLNA** MCQ.
- Theories are early urethral obstruction, a lateral plate mesoderm defect, an intrinsic urinary tract defect and a yolk sac defect.

> **KEY POINT**
> Weakness of the obstruction theory: most have **no obstruction after birth**. The tract is **dilated, collagen-rich and largely non-obstructed**, and this drives management.

2 · Genitourinary pathology

- **Kidneys**: dysplasia in ~**50%**, worst with outlet obstruction without urachal decompression. Dilation does not correlate with dysplasia; **infection, not obstruction, is the main threat**.
- **Ureters**: the distal ureter is worse, so the **proximal ureter is preserved** in reconstruction. Vesicoureteric reflux (VUR) occurs in ~**75%**. The ureter is a **failed pump, not a blocked pipe**.
- **Bladder**: massive, thick but smooth, with a urachal pseudodiverticulum; the urachus is patent at birth in **25–30%**. It has normal compliance, large capacity and poor sensation; about half void normally but may deteriorate.
- **Prostatic hypoplasia** dilates the posterior urethra and causes ejaculatory failure. **Urethral atresia** is lethal unless the urachus is patent. **Megalourethra** is scaphoid (spongiosum only, milder) or fusiform (cavernosa too, more severe).
- **Testes** lie over the iliac vessels. Infant germ cells are normal, so the abdominal environment destroys later potential, hence early orchidopexy.

3 · Extragenitourinary anomalies

About **75%** have them, and nearly **half premature**. Pulmonary problems (>50% of survivors) include hypoplasia, pneumothorax and pneumonia; cardiac (~10%), gastrointestinal (≥30%, malrotation) and orthopaedic (up to 65%, talipes, hip dysplasia, knee dimples) anomalies are common.

KEY POINT

Two mechanisms: **oligohydramnios** (lung hypoplasia, limb deformity, knee dimples) and **absent abdominal wall pressure** (weak cough, atelectasis, constipation).

4 · Presentation and spectrum

- **Prenatal** appearance mimics posterior urethral valves: massive bladder and hydroureteronephrosis. Dilation does **not** predict function.

Woodard category	Kidneys / lungs	Course	Urological management
I	Dysplasia; **pulmonary hypoplasia**; oligohydramnios, urethral atresia	Stillbirth or early death (~20% perinatal mortality overall)	Supportive; simple bladder drainage
II	Full triad; minimal or unilateral dysplasia; no lung hypoplasia	Risk of progressive renal insufficiency	Individualised — **controversial group**
III	Incomplete or mild; no dysplasia; stable	Majority; good	Surveillance; surgery for UTI or deterioration; orchidopexy in year 1

- The abdominal wall is not a sorting criterion. **Pseudo-prune** (partial flaccidity, unilateral undescended testis) is diagnosed late and has higher late renal failure.

5 · Evaluation

ALGORITHM

1. Team; **heart and lungs first** (chest X-ray).
2. Creatinine trend; the day-1 value is maternal.
3. Renal–bladder ultrasound.
4. Prophylaxis and circumcision; then selective imaging.

- A voiding cystourethrogram is done only for renal insufficiency or suspected outlet obstruction, under antibiotic cover. A **DMSA** scan is done at **4–6 weeks**; MAG3 renograms are unreliable with massive dilation. Outlet obstruction is drained by a percutaneous suprapubic tube.

6 · Management philosophy

	Conservative	Early reconstruction
Premise	Dilation is non-obstructive and may improve	Stasis, reflux and infection drive renal loss
Operate when	Proven obstruction, intractable infection	Progressive dilation, high-grade VUR, recurrent upper tract infection, declining function
Best for	Category III	Selected category II

- Delay major reconstruction to **≥3–6 months** for the lungs. Single-stage comprehensive surgery adds no morbidity.

7 · Urinary tract surgery

- **Blocksom vesicostomy** (sepsis, renal failure, atresia): make the stoma larger than usual, as it tends to stenose. Avoid proximal ureterostomy.
- **Ureteroplasty**: excise the distal ureter, taper the upper part and reimplant. Reduction cystoplasty is reserved for a large urachal diverticulum, as capacity recurs; clean intermittent catheterisation empties better. Circumcision reduces UTI.

8 · Orchidopexy

- A broad mesorchium defeats the inguinal approach. Do a **bilateral transabdominal orchidopexy at ~6 months**, often with other surgery. Later, a **staged Fowler–Stephens** procedure is preferred.

9 · Abdominoplasty

- Cosmetic gain is certain; functional gains (cough, defecation, emptying) are plausible. Operate from ~6 months with other surgery.

Technique	Incision / umbilicus	Key step / note
Randolph	Transverse, 12th rib to 12th rib	Excision of the lower wall; lateral bulge persists
Ehrlich	Vertical; umbilicus on a pedicle	Vest-over-pants advancement
Monfort	Vertical ellipse; umbilicus in situ	Central fascial plate with lateral flaps; **transperitoneal**, so ideal with reconstruction
Dénes	Fusiform	Modified Ehrlich
Furness	Midline, limited	Plication; peritoneum not opened

ALGORITHM

1. **Monfort**
2. Skin ellipse, umbilicus spared; incise the central plate at the lateral rectus borders.
3. Do the genitourinary surgery.
4. Suture plate edges to scored lateral peritoneum.
5. Overlap lateral flaps over drains.

10 · Long-term outcome

- Creatinine **<0.7 mg/dL at 1 year** predicts stable function. Up to **30%** develop chronic renal failure, driven by pyelonephritis, dysplasia and polyuria. Transplant outcomes are equal.
- **Bladder**: timed and double voiding; if these fail, use CIC.
- **Fertility**: puberty and erections are normal, but **azoospermia means no spontaneous paternity**; sperm retrieval with ICSI is used.

KEY POINT

Long-term care is a contest against **stasis**: lowering residual urine and infection protects the kidneys, while dysplasia and polyuria set a ceiling surgery cannot raise.

45 Posterior Urethral Valves

1 · Definition and classification

- **Posterior urethral valves (PUV)** are the commonest cause of bladder outlet obstruction in infant boys. Ablation relieves obstruction but **not the embryological injury** to bladder and kidney, so care is lifelong.
- Lower urinary tract obstruction (LUTO) affects ~2.2/10,000 births; PUV is commonest (~1.4/10,000), then urethral atresia and prune-belly.

P1 Embryology of Posterior Urethral Valves. — Young 1919; types I–III and origin; COPUM (one membrane torn by catheter); genetics: chr 11 duplications, CNVs (worse renal outcome), TBX5, PTK7

Not in the paper answer:

- Type III resembles a congenital stricture; it is contentious but accepted.

Young type	Anatomy	Status
I (95%)	Leaflets from verumontanum fuse anteriorly just proximal to external sphincter	Obstructing; the lesion treated
II	Folds from verumontanum to bladder neck	Non-obstructing; probably not real
III (5–10%)	Annular membrane, central aperture (persistent urogenital membrane)	Obstructing

- Copy number variant burden predicts higher nadir creatinine and CKD stage 5.

2 · Pathophysiology

- Obstruction causes detrusor hypertrophy, then **fibrosis replaces smooth muscle**. The bladder is the **driver**: renal dysfunction, reflux (VUR) and voiding dysfunction act through it.
- The voiding cystourethrogram (VCUG) triad is a **dilated posterior urethra, hypertrophied bladder neck and flattened verumontanum**.
- Ureters dilate from **pressure transmission, VUR (up to 70%) and polyuria**, then lose peristalsis, causing stasis and infection.
- Renal injury is **obstructive uropathy** (partly reversible) plus **dysplasia** (fixed at birth, which sets the ceiling). Tubular injury causes **nephrogenic polyuria** and post-obstructive diuresis.
- Angiotensin II induces TGF-β1 and fibrosis; ACE inhibitors may help selected children.

Organ / lesion	Effect	Natural history
Lung — hypoplasia	Respiratory failure	Commonest cause of neonatal death
Kidney — obstructive uropathy	Reversible insufficiency	Improves; may recur with bladder dysfunction
Kidney — dysplasia	Irreversible insufficiency	Progressive failure, hypertension
Kidney — tubular injury	Salt and water wasting	Nephrogenic diabetes insipidus; worsens with age
Bladder	Poor sensation, hypercontractility, low compliance; later myogenic failure	Lifelong, changes with age

Organ / lesion	Effect	Natural history
Ureters	Poor coaptation, stasis	Most retain hydronephrosis — **not** a sign of residual obstruction

"Pop-off" mechanisms

- **VURD** (unilateral high-grade VUR into a non-functioning dysplastic kidney) was thought protective. This is **refuted**: most with normal creatinine at 2 years lose it by 8–10 years, and urodynamics are worse. Urinoma, ascites and diverticula are debated but managed the same way.

> **KEY POINT**
> Every pop-off is false reassurance: normal creatinine in a growing infant hides an abnormal GFR. Follow VURD as closely as any PUV child.

3 · Diagnosis

Antenatal

- Ultrasound shows a **thick distended bladder, bilateral hydroureteronephrosis, "keyhole sign" and oligohydramnios**; an echogenic or cystic cortex means dysplasia. It is only ~50% accurate for the cause.
- About half are diagnosed antenatally, but this has **not** improved renal outcome. A male fetus with these findings needs postnatal US and a **VCUG before discharge**.

Postnatal

- **US** comes first: the wall is thick if **>3 mm distended / >5 mm empty**.
- **VCUG** is definitive, showing a dilated elongated posterior urethra with **abrupt funnelling** at the valve; about half have high-grade VUR. Use a 5–8 Fr feeding tube and **withdraw the catheter during voiding**.
- A **MAG3** renogram is done at **2–3 months**. Creatinine in the first 48 h is maternal; follow the trend to the **nadir** in year 1.

4 · Fetal intervention

- Decisions hinge on **oligohydramnios, hydronephrosis degree and laterality, and gestational age**. Favourable features are oligo- (not an-)hydramnios, normal karyotype, no cortical cysts and favourable urine.

Fetal urine (after 20 weeks)	Favourable
Sodium / chloride	**<100 / <90 mEq/L**
Osmolality	**<200 mOsm/L**
β2-microglobulin	**<6 mg/L** (most sensitive)

- **Triple serial sampling** 48–72 h apart beats one sample, though prediction was poor on systematic review.
- A **vesicoamniotic shunt** may prevent **pulmonary hypoplasia** but gives **no renal benefit**. Fetal cystoscopic laser ablation is no better. Both carry **40–50%** pregnancy loss.

KEY POINT

Fetal intervention can rescue the lung, not the kidney: fluid deficiency is still in progress, but dysplasia is already established.

5 · Neonatal presentation and initial management

- **Pulmonary hypoplasia** is the commonest early death and the usual cause of delayed surgery.

ALGORITHM

1. Stabilise the airway.
2. Pass a **5–7 Fr feeding tube**. If it curls: coudé, curved stylet, or rectal finger.
3. Confirm on US; minimal balloon fill.
4. Creatinine, electrolytes, fluids.
5. US, then VCUG; antibiotic prophylaxis.
6. Ablate when stable.

- **Post-obstructive diuresis** needs monitoring of output, weight and electrolytes with IV replacement. **Hyponatraemia** is transient pseudohypoaldosteronism, treated with sodium.
- **Urinoma or ascites** (3–10%) resolves with bladder drainage.
- **Delayed presentation** (UTI, incontinence, renal failure) is common: over 40% present after 6 months, and late does not mean milder.

6 · Surgery

Primary valve ablation (preferred)

- Ablation restores flow and **normal bladder cycling**, superior to passive diversion. Use a **7.5–9 Fr cystoscope** with holmium laser or Bugbee.
- Incise at **5 and 7 o'clock, with or without 12 o'clock**. Valves are thin and avascular, so **incise, don't resect**.
- For a small urethra, use vesicostomy or **PADUA** (progressive urethral dilation with up-sized tubes). Leave a catheter ≥24 h and **repeat VCUG within 1 month**.

Complication	Cause / prevention	Management
Urethral stricture	Over-resection, hot loop, oversized instrument	Dilation / urethrotomy; urethroplasty if dense
Meatal injury	Resectoscope through small meatus	Smaller scope / PADUA
Residual valve	Incomplete incision	Persistent funnel on VCUG: re-ablate
Sphincter injury	Ablating distal to valves	Incontinence — precise incision

- **Bladder neck incision** gives no benefit and risks retrograde ejaculation; it is not recommended.

Vesicostomy

- It is used for a very small or premature infant, failed decompression, or persistent deterioration. It **does not defunctionalise** the bladder and matches ablation in final outcome.

P4 Vesicostomy. — Blocksom: 2 cm transverse incision midway, dome (not anterior wall) to skin to stop prolapse; indications, complications, closure

Not in the paper answer:

- The urachus is excised with a small dome cuff; using the dome keeps the posterior wall taut.

Upper tract diversion, circumcision and nephrectomy

- Upper tract diversion gives **no renal protection**; use it only for deterioration **despite full lower tract drainage**.
- UTI risk is 50–60% in PUV; circumcision cuts it by over 80%, so offer it to all.
- VURD nephroureterectomy is rare; **keep the ureter** for augmentation.

7 · VUR in PUV

- VUR occurs in 50–80% and is **secondary**. It resolves after ablation in 25–40% and does not predict renal outcome.
- Treat the **bladder**. Reimplant only for recurrent febrile UTI despite circumcision and optimal bladder care.

> **KEY POINT**
> VUR in valves is a pressure gauge, not a lesion: fix the pressure, not the reading.

8 · Bladder dysfunction and valve bladder syndrome

P2 Valve bladder: management — Mitchell definition; polyuria, poor sensation, residual; age pattern; continence ~5.5 y; ladder: behavioural, then oxybutynin 0.1 mg/kg bd / α-blocker, then CIC, then overnight drainage 7–10 h, then Mitrofanoff, then ureterocystoplasty

Not in the paper answer:

- Stop oxybutynin if residual and capacity rise, as myogenic failure can follow.

> **ALGORITHM**
> 1. High-pressure voiding causes fibrosis and weaker contraction.
> 2. Post-void residual (PVR) rises; nephrogenic polyuria adds volume.
> 3. Upper tracts store urine and refill the bladder, so it is never empty.
> 4. Pressure is back-transmitted; renal function declines, causing more polyuria.

- Continence comes at ~5.5 vs 2.9 years. Do **uroflow and PVR** in every toilet-trained boy; new incontinence after dryness is a red flag.
- An α-blocker suits early bladder-neck obstruction and an antimuscarinic the overactive wall; both harm once the bladder is hypocontractile. Do **videourodynamics** serially.

9 · Prognosis for renal function

- Lifetime end-stage renal disease (ESRD) is **20–50%**. Risks are dysplasia, **year-1 nadir creatinine**, recurrent UTI and bladder dysfunction.

Nadir creatinine (1 year)	ESRD risk
<0.8 mg/dL	Minimal (best single predictor)
>1.2 mg/dL	High
<0.4 / 0.4–0.69 / 0.7–0.99 / >1.0 mg/dL	**0% / 2% / 27% / 100%**

- **0.7 mg/dL is the watershed**, where risk leaps. On US, echogenicity, cysts and a smaller renal parenchymal area predict a worse outcome.

10 · CKD, ESRD and transplantation see Ch 61

- Do **videourodynamics in every candidate** and optimise with CIC or overnight drainage.
- Graft and patient survival are equal if the bladder is managed; late graft loss follows bladder dysfunction. Augmentation can be done before or after transplant.

11 · Long-term follow-up, sexual function and fertility

- Transition to adult urologists who know valves. Adult LUTS are 2–3 times commoner; erectile function and fertility are similar overall but worse with incontinence or renal insufficiency.
- Lifelong checks are blood pressure, creatinine, urine protein, US, and uroflow with PVR.

12 · Related urethral lesions

P2 Anterior urethral valve — 25–30× rarer than PUV; diverticulum; penile bulge; vesicostomy if premature, else endoscopic ablation; open repair of large diverticulum; 80% bladder dysfunction, 78% normal renal function; 25-fold risk with azotaemia+VUR+UTI

- **Urethral atresia** needs decompression to survive; treat with vesicostomy, then PADUA.
- **Urethral duplication** (Effmann I–III): the **ventral** channel has the sphincter, so it is the one moved to the glans.
- **Urethrorrhagia** in peripubertal boys resolves in up to 92%; treat voiding dysfunction and reserve cystoscopy for atypical cases.
- Anorectal malformation often has a rectourinary fistula; catheterise before PSARP.

13 · Landmark trials

Trial	Design	Result	Take-home
PLUTO	RCT: vesicoamniotic shunt vs conservative in LUTO	Trend to better 28-day survival; poor long-term survival and renal function in both	Shunt may help lungs, not kidneys
Circumcision RCT	Circumcision and prophylaxis vs prophylaxis	Febrile UTI 3% vs 49% over 2 years	Circumcise boys with PUV
Oxybutynin RCT	Oxybutynin to toilet training vs observation post-ablation	Better hydronephrosis/VUR; no change in creatinine, UTI, continence	Selective, monitored use

46 Neuromuscular Dysfunction of the Lower Urinary Tract in Children

1 · Definition and causes

- **Neurogenic lower urinary tract (LUT) dysfunction** is bladder and/or sphincter dysfunction from aberrant innervation.
- The commonest cause is **neural tube defect (NTD)**, mainly spina bifida (SB). Survival is about 90% to 10 years, so **renal failure** is a leading late morbidity; **10%** of SB deaths are urological. Clean intermittent catheterisation (CIC, Lapides 1971) was the pivotal advance.

Congenital	Acquired
Myelomeningocele; occult NTD (lipomeningocele, tethered cord); sacral agenesis; anorectal malformation (ARM)	Brain: cerebral palsy (CP), tumours, infarcts. Cord: trauma, transverse myelitis. Pelvis: extensive surgery

2 · Myelodysplasia (NTD)

P1 Neurogenic bladder in myelodysplasia — lesion levels; detrusor × outlet patterns; EAU work-up; McGuire risk; early CIC outcomes

Not in the paper answer:

- Trospium (a quaternary amine) has few central effects.

Classification and prevention

- **Myelomeningocele** is open, commonest and most severe; **SB occulta** is closed and usually benign.
- The neural tube closes by **day 35**. **Folic acid** cuts NTDs by 50–70%: all women of child-bearing age take **400 µg/day** from 4 weeks before conception, and women with SB take **4–5 mg/day**.
- Recurrence is **5%** after one affected child. Almost all open NTDs have **Chiari II**, and most have hydrocephalus.
- **LUT function is not predictable from bony level or the legs**, so urodynamics defines every bladder.

> **KEY POINT**
> Management of Myelomeningocele Study (**MOMS**): prenatal repair (<26 weeks) reduced death or shunt need (RR 0.70) and improved motor and cognitive scores at 30 months, at the cost of prematurity, uterine-scar thinning or dehiscence, and more re-tethering.

Pathophysiology

- In **detrusor–sphincter dyssynergia (DSD)**, the sphincter fails to relax during detrusor contraction; it is diagnosed by electromyography.

> **KEY POINT**
> Every pattern is detrusor (over/under) × outlet (competent/incompetent/dyssynergic). Dangerous: **overactive or poorly compliant detrusor against a closed or dyssynergic outlet**, where storage pressure rises and causes reflux and hydroureteronephrosis. An incompetent outlet "pops off": the child is wet but the kidneys are safe.

Newborn evaluation

ALGORITHM

1. Before discharge: renal–bladder US and post-void residual (need for CIC?).
2. After week 1: serum creatinine.
3. Once stable: urodynamics, **safe from ≈3 months**.
4. Cystography: ideally videourodynamics, otherwise VCUG.
5. Optional baseline DMSA scan (EAU).
6. US at 6, 12, 18 and 24 months, then yearly.

- Guidelines come from ICCS 2012, BAPU 2016 (most selective), SBA 2018 and EAU 2022. In the UMPIRE newborn cohort, dilating VUR was **7%**.
- **Cystatin C** beats creatinine, because low muscle mass hides a falling GFR. High-grade VUR acts as a pop-off, so urodynamic pressures underestimate risk.

Urodynamic risk factors

Category	Definition
Hostile / high	Detrusor leak point pressure (DLPP) or end-filling pressure **≥40 cmH_2O**; NDO with DSD (DSD removed in UMPIRE revision)
Intermediate	NDO, reduced compliance, 25–39 cmH_2O
Abnormal but safe	<25 cmH_2O
Normal	Normal capacity, compliance >15 mL/cmH_2O, no NDO/DSD, minimal residual

- Poor compliance independently predicts renal injury.

Tethered cord

- Growth stretches a low-lying cord, so gait, sensation and bladder deteriorate. Suspect it with new wetting, rising pressure, new UTIs, worsening scoliosis or back pain. Do urodynamics **before untethering and 3–4 months after**.

3 · Proactive (early) management

Priorities in order are **kidneys, then continence, then UTI avoidance, then sexual function and fertility**.

- **Early CIC and an antimuscarinic** (Bauer) treat hostile dynamics. All paediatric societies endorse this; there is **no benefit if the child empties spontaneously**. If hydronephrosis persists, add overnight drainage.
- Give **no** antibiotic prophylaxis on CIC; treat bacteriuria only if symptomatic.

Outcome	Expectant	Early CIC + antimuscarinic
Hydroureteronephrosis	18–80%	0–15%
Renal scarring	27–50%	4–12%
End-stage renal disease (ESRD)	≈30%	0–1.6%
Augmentation	—	18–24% fewer

KEY POINT

Proactive won because it treats the cause (pressure, failure to empty) before the consequence (reflux, dilatation, scarring, renal failure).

4 · Management ladder

ALGORITHM

1. CIC and antimuscarinic, with or without a β3-agonist.
2. If that fails: intradetrusor botulinum toxin.
3. If CIC is impossible: vesicostomy as a bridge.
4. Then augmentation, with or without a catheterisable channel and an outlet procedure see Ch 48.

Pharmacotherapy

- **Antimuscarinics** are first-line. FDA-licensed drugs are **oxybutynin** (≥5 y; 0.1–0.6 mg/kg/day, max 5 mg/dose; oral, patch or intravesical), **solifenacin** (≥2 y), **fesoterodine** (≥6 y, >25 kg) and **mirabegron** (≥3 y; may cause hypertension).
- α-Sympathomimetics for low-outlet incontinence are unlicensed and help a minority.

Botulinum toxin A

- Under general anaesthesia, inject **≈10 U/kg (max 360 U)** into about 30 sites, **sparing the trigone**. Continence is 65–87%, lasting 6–10 months, and repeats stay effective. Poor pre-existing compliance predicts poor response.

Surgery

- **Vesicostomy** is a temporary bridge when CIC is unmanageable. **Augmentation** (ileum preferred) treats the refractory bladder, with a channel or outlet procedure as needed see Ch 48. After ileum, check B12 yearly.

VUR in the neurogenic bladder

- VUR is secondary, and treating the bladder resolves 30–55%. **Avoid Credé** with a reactive sphincter, because it worsens VUR.
- Antireflux surgery is for breakthrough febrile UTI despite low pressure. At augmentation, **reimplant grade ≥III**.

5 · Bowel management and latex allergy

- Neurogenic bowel is managed stepwise with diet, laxatives and transanal irrigation, then a **Malone antegrade continence enema (ACE)**, often at augmentation.
- Latex sensitisation risks intraoperative anaphylaxis, so care is latex-free **from birth**.

6 · Sexual function and transition

- Function is **inversely proportional to neurological level**. Central precocious puberty affects about half of girls.
- **Males** have erectile dysfunction in about 75%; **sildenafil** works in about 80%. **Females** conceive in 70–80%; avoid the combined pill and IUD. Offer a structured transition.

7 · Other causes

- **Sacral agenesis** is linked to maternal diabetes; look for flat buttocks and a short gluteal cleft. Level does not predict the bladder, and about a third present late.
- **Spinal cord injury**: CIC during spinal shock; a **suprasacral lesion later gives NDO with DSD**. **Autonomic dysreflexia** (lesion **≥T6**) causes severe hypertension with bladder filling, so monitor blood pressure during urodynamics.
- **Transverse myelitis** causes acute retention in over 95%; **late CIC leads to poorer compliance**, so do early urodynamics and CIC.
- **Cerebral palsy** gives coordinated overactivity; upper-tract risk is **<5%**, so urodynamics is only for failed conservative care or recurrent UTI.
- **ARM** has a cord anomaly in 18–50%, so do spinal US before 3 months; neuropathy comes from the cord, not the PSARP. Pelvic surgery can injure pelvic nerves.

KEY POINT

Lesion level predicts pattern. **Suprapontine** (CP) gives coordinated overactivity and low risk. **Suprasacral cord** (SCI, myelitis, tethering) gives NDO with DSD and high risk. **Sacral/peripheral** gives underactivity. Myelodysplasia is mixed and unpredictable.

8 · Landmark trials

Trial	Design	Result	Take-home
MOMS	RCT, prenatal vs postnatal repair	Less shunting, better motor score; more prematurity, uterine dehiscence	Fetal repair in selected centres
McGuire 1981	Myelodysplasia cohort	DLPP >40 cmH_2O led to upper-tract damage	40 cmH_2O = safety line
Sildenafil in SB	Double-blind placebo RCT	≈80% improved erections	First-line for ED

47 Functional Disorders of the Bladder and Bowel in Children

1 · Normal bladder and bowel function

P1 Describe neurophysiology of lower urinary tract and various types of voiding dysfunction in children — innervation, PAG–PMC control, maturation, ICCS functional types, flow patterns, management principles

Not in the paper answer:

- The International Children's Continence Society (ICCS) now uses expected bladder capacity = (age + 1) × 30 mL; the paper's (age + 2) is the older formula†.
- In **infancy**, voiding pressures are often **>100 cm H2O** and interrupted voiding is normal; both are gone by **~2 yr**. Dyscoordination later is **dysfunction, not immaturity**.
- Late toilet training is linked to incontinence, enuresis and urinary tract infection (UTI).

> **KEY POINT**
> Bladder and bowel are one unit, sharing pelvic space and afferents. A loaded rectum causes overactivity and poor emptying, so treating constipation improves lower urinary tract symptoms (LUTS).

2 · Definitions — ICCS

Bladder and bowel dysfunction (BBD) is lower urinary tract (LUT) dysfunction plus constipation and/or faecal incontinence in a neurologically normal child.

Group	Symptom	ICCS definition
Storage	Increased frequency	**≥8 voids/day**
	Decreased frequency	**≤3 voids/day**
	Urgency	Sudden immediate need to void
	Nocturia	Waking at night to void
Voiding	Hesitancy	Difficulty initiating when ready
	Straining	Valsalva to initiate and maintain
	Intermittency	Non-continuous stream
	Retention	Cannot void despite effort; distended bladder
Other	Holding manoeuvres; incomplete emptying; recurrent UTI	Incomplete-emptying sensation counts even with minimal post-void residual (PVR); recurrent UTI is itself a BBD symptom

- **Daytime urinary incontinence (DUI)** is leakage while awake at age **≥5 yr**. **Enuresis** is intermittent leakage while asleep at ≥5 yr: **monosymptomatic (MNE)** has no other LUTS, **non-monosymptomatic (NMNE)** has some.
- **Faecal incontinence** is mostly **retentive** (overflow); ~1 in 5 is non-retentive.
- † **Expected bladder capacity (EBC)** = (age in years + 1) × 30 mL (ICCS, to ~12 yr).

Daytime LUT conditions

Condition	Definition / mechanism	Key features
Overactive bladder (OAB)	Urgency, with or without frequency, nocturia, incontinence; no UTI or other pathology	Urgency incontinence, holding manoeuvres
Dysfunctional voiding	Habitual sphincter/pelvic-floor contraction **during voiding**, neurologically intact; staccato flow	Learned; high PVR, UTI, secondary vesicoureteral reflux (VUR)
Underactive bladder	Raises abdominal pressure to initiate, maintain or complete voiding	Infrequent, interrupted flow, PVR
Voiding postponement	† Habitual deferral with holding until urgency incontinence	† Low frequency; oppositional defiant disorder
Giggle incontinence	Extensive emptying during or after laughing; LUT otherwise normal	Distinguish from laugh-provoked OAB leak and vaginal reflux
Vaginal reflux	Vagina fills retrograde during voiding	Prepubertal girl; small leak just after voiding
Extraordinary daytime frequency	† ≥ hourly small voids, no incontinence or nocturia	† Benign, self-limiting; exclude UTI, diabetes
Hinman syndrome (non-neurogenic neurogenic bladder)	End stage of dysfunctional voiding	Trabeculation, VUR, hydronephrosis, renal damage, hypertension

KEY POINT

Trap: dysfunctional voiding (neurologically intact) is not detrusor–sphincter dyssynergia (DSD), which needs a neurological lesion. Urgency incontinence is not always OAB: a voiding postponer also leaks with urge, and anticholinergics help briefly, then worsen emptying.

3 · Clinical presentations and associations

- **Acute retention** (peak 2–4 yr) follows genital inflammation, instrumentation or **faecal impaction**; try **analgesia, sitz bath and/or enema before a catheter**.
- **Functional constipation** affects ~14% and is driven by a painful-stool **withholding cycle**. Exclude organic causes such as anorectal malformation (ARM), Hirschsprung disease and spinal causes.
- **ADHD, autism and intellectual disability** have the highest BBD rates and respond worse.

4 · Evaluation

*Evaluation mainly **excludes organic disease**; avoid over-investigation.*

Warning signs

Red flags are rectal bleeding, weight loss or vomiting (organic disease); poor growth or hypertension (chronic kidney disease, CKD); a lumbosacral skin lesion or abnormal leg neurology (spinal MRI); an abnormal anus (ARM); low specific gravity (SG) or glycosuria (diabetes insipidus, DI, or diabetes mellitus); and constant dribbling in a girl (ectopic ureter).

- Ask about bowels in **every** child with LUTS, and use a **3-day bladder diary** and a **bowel diary**. Bristol types 1–2 indicate constipation.

- A **gush on separating the labia** means vaginal reflux. Routine rectal examination is **not** needed.

Investigations (stepwise, non-invasive)

- **ICCS response**: complete 100%, partial 50–99%, none <50%.
- **Urinalysis**: SG >1.005 makes DI unlikely. Persistent proteinuria needs a first-morning protein:creatinine ratio (>**0.2 mg/mg** is abnormal).
- Measure **PVR** within 5 min of voiding at 50–115% EBC.
- **Uroflow** (3 flows, >50% EBC): bell is normal, tower suggests OAB, staccato dysfunctional voiding, interrupted underactive bladder, and plateau obstruction.
- Pelvic floor **EMG** active during voiding means dysfunctional voiding; an **EMG lag time <2 s** means detrusor overactivity. **Video urodynamics** is only for treatment failure or suspected obstruction or upper-tract risk.

Bowel — Rome IV functional constipation (child ≥4 yr)

ALGORITHM

1. ≥2 of, ≥1/week for ≥1 month, irritable bowel syndrome (IBS) criteria not met:
2. ≤2 defecations in toilet/week; ≥1 faecal incontinence/week; retentive posturing/withholding; painful or hard stools; large rectal faecal mass; large-diameter stools that may block the toilet.

- Pain is **not** a Rome IV criterion.
- No routine labs or **plain radiograph**. A rectal diameter **≥30 mm** on US suggests constipation (European Association of Urology/European Society for Paediatric Urology, EAU/ESPU).
- If refractory, use a transit study, contrast enema and **anorectal manometry** (absent RAIR needs rectal biopsy for Hirschsprung).

5 · Management of daytime incontinence

ALGORITHM

1. History, exam, urinalysis, PVR, with or without uroflow.
2. Exclude neurogenic causes, diabetes, CKD and anatomical causes.
3. Optimise neurodevelopmental disorders; **treat constipation first**.
4. If wetting is only at night, treat as MNE.
5. Otherwise, **standard urotherapy** (≥4 wk); re-evaluate.
6. Then **specific urotherapy** by diagnosis.
7. If refractory: botulinum toxin or neuromodulation.

Standard urotherapy

- Give information, fluid and diet advice and diaries. Sit with hips and knees at ~90° and feet supported, and use **timed voiding** 6–7 times a day. **Reward adherence, not dryness**. Urotherapy cures DUI in more than half.

Dysfunctional voiding

- Diagnose by **staccato flow with active pelvic floor EMG**.

Renal risk	Management
High (hydronephrosis, VUR, rising creatinine)	Video urodynamics; lower storage pressure; clean intermittent catheterisation (CIC) past the sphincter
Low	Urotherapy, double voiding, treat constipation, short-course prophylaxis for recurrent UTI

- **Biofeedback is the first-line specific therapy**. **α-blockers** (doxazosin) lower PVR. If refractory, give **intrasphincteric onabotulinumtoxinA** 50–300 units.

Overactive bladder

Use drugs only after urotherapy and bowel management.

Agent	Dose / point
Oxybutynin (non-selective)	Immediate release **0.3–0.6 mg/kg/day** in 2–3 doses (max 15 mg); extended release (ER) 5–20 mg/day; patch 3.9 mg/day. Only US-approved drug for paediatric OAB; causes dry mouth and constipation.
Tolterodine	1–2 mg twice daily or ER 2–4 mg; ER randomised controlled trial (RCT) negative
Solifenacin (M3-selective)	2.5–10 mg daily; positive phase III RCTs (LION, LEOPARD)
Trospium (quaternary amine)	Does not cross the blood–brain barrier
Mirabegron (β3-agonist)	25–50 mg if >35 kg; approved ≥3 yr for **neurogenic** detrusor overactivity only; safe add-on

- Check PVR and stools before and during antimuscarinics.
- **Intradetrusor onabotulinumtoxinA** (refractory): **5 units/kg, max 100 units**; it lasts ~12 months.

Neuromodulation options are parasacral TENS over S2–S3 (beat oxybutynin, but a sham RCT was negative), the weaker tibial nerve stimulation, and off-label sacral nerve stimulation (SNS), which helps bladder and bowel.

Other conditions

- **Underactive bladder**: timed and double voiding and biofeedback; CIC if PVR is high.
- **Postponement**: timed voiding and treat comorbidity; † avoid anticholinergics.
- **Giggle**: **methylphenidate** or **biofeedback**.
- **Vaginal reflux**: void with legs abducted (sit backwards on the toilet). **Extraordinary frequency**: † reassure.

6 · Nocturnal enuresis

P1 P3 Nocturnal enuresis / Enuresis — ICCS definitions, primary vs secondary (6 months dry), triad, sleep apnoea, evaluation, alarm vs desmopressin, second line

Not in the paper answer:

- Prevalence is **5–10% at 7 yr**; ~15% resolve each year, and ~2% persist into adulthood.
- The **triad** is nocturnal polyuria (relative antidiuretic hormone deficiency), reduced nocturnal capacity or overactivity, and failure to arouse.
- In **NMNE**, treat bothersome daytime LUTS and constipation first. In **MNE**, treat from **~6 yr**.

	Enuresis alarm	Desmopressin
Use	Nightly; the parent wakes the child; continue until **14 consecutive dry nights**	**0.2–0.4 mg** tablet or 120–240 µg melt ~1 h before bed; stop if no effect after ~1 week
Response	50–70%; slow	Immediate; rule of thirds; best with nocturnal polyuria
Durability	**Durable**; overlearning lowers relapse	**Not durable**; drug holidays or special nights
Safety	Nil	Hyponatraemia: † no fluids from 1 h before dose

ALGORITHM

1. MNE at ≥6 yr: start the alarm or desmopressin.
2. If it fails, check use, then switch or **combine**.
3. Re-evaluate: hidden daytime LUTS, constipation, sleep-disordered breathing, neurodevelopmental disorder; uroflow and PVR.
4. Then **add an anticholinergic** to desmopressin.
5. If refractory: imipramine, a β3-agonist or specialist drugs.

- **Imipramine** 25–50 mg is **cardiotoxic** (overdose, long QT), so do an electrocardiogram if there is syncope, palpitations or a family history of sudden death.

7 · Functional constipation — treatment

P4 Management of Defecation Disorders resulting into LUTS — why bowel causes LUTS, Rome IV, bowel first, disimpaction/maintenance PEG, biofeedback, irrigation, ACE, toxin, neuromodulation

Not in the paper answer:

- Daily enemas for 3–6 days are an alternative to PEG.

ALGORITHM

1. If impacted, **disimpact** with oral polyethylene glycol (PEG) or enemas.
2. Give education, a diary, toilet sitting and **maintenance PEG**.
3. Review at 2 wk; if failing, check adherence, dose, impaction, mental health.
4. Continue for **≥2 months**. Wean when ≥1 month symptom-free, with ≥3 stools/week and Rome IV not met.
5. Relapse: reinstate, or refer to gastroenterology.

Laxative	Dose
PEG 3350 — disimpaction	**1–1.5 g/kg/day × 3–6 days** (>3 yr)
PEG 3350 — maintenance	**0.2–0.8 g/kg/day**, start 0.4 (>6 months)
Lactulose — maintenance	1–2 g/kg once or twice daily (safe in infants)

- **Titrate** PEG: increase it if there is no stool in 24 h, and add a **stimulant** (senna, bisacodyl) if watery accidents occur. There is **no dependence**.
- Refractory cases may need biofeedback, transanal irrigation or SNS. Before an antegrade continence enema (**ACE**), do **colonic manometry**, because diffuse dysmotility may need resection (see Ch 48).

8 · Recurrent UTI and BBD

- In RIVUR/CUTIE, recurrence was **BBD + VUR 51% > BBD alone 35% > VUR alone 20%**.

- Secondary VUR often resolves once BBD is treated.
- **EAU/ESPU**: exclude BBD in every toilet-trained child with febrile or recurrent UTI, and give prophylaxis during BBD treatment. The **AUA** advises prophylaxis for BBD with VUR.

9 · Landmark trials

Trial	Design	Result	Take-home
RIVUR + CUTIE (BBD analysis)	Prophylaxis vs placebo, <6 yr after febrile UTI, with or without VUR	BBD + VUR highest recurrence; BBD alone > VUR alone	Treat BBD; prophylaxis for BBD + VUR
LION / LEOPARD	Phase III placebo RCTs, solifenacin	Improved all efficacy measures; mild–moderate side effects	Best paediatric antimuscarinic evidence

48 Lower Urinary Tract Reconstruction in Children

1 · Goals and principles

- The priorities, in order, are: **protect the kidneys, then prevent urinary tract infection (UTI), then achieve continence**.
- The indication is a hostile bladder (poor compliance, high pressure, poor emptying) despite medical care, mostly in spina bifida; also exstrophy, posterior urethral valves and bilateral ectopic ureters.
- **Clean intermittent catheterisation (CIC)** is the key enabler, and the main determinant of outcome is **patient and family commitment**.

> **KEY POINT**
> Every reconstructive step trades storage for emptying. A continent reservoir that is not catheterised causes hydronephrosis, sepsis and perforation, so incontinence can be "protective". CIC and motivation are prerequisites.

2 · Patient selection and preoperative evaluation

- **Storage**: a chronic filling pressure **>40 cm H2O** causes hydroureteronephrosis, a falling glomerular filtration rate (GFR) and acquired vesicoureteral reflux (VUR). Start anticholinergics and CIC; persistent poor compliance needs **augmentation**.
- **Outlet**: resistance must exceed bladder pressure during all activity; a deficient outlet needs a procedure.
- **Emptying**: nothing predicts voiding after augmentation, so every patient practises CIC beforehand.

Urodynamic finding	What it decides
Capacity and fluoroscopic bladder shape	Length of bowel segment
Detrusor leak point pressure (DLPP), end-filling pressure, open bladder neck	Need for a bladder neck procedure
VUR with a history of UTI	Reimplantation
Sphincter electromyography (EMG)	Dyssynergia

- Fill at **≤10% of predicted capacity per minute**, because fast filling falsely lowers compliance.
- Every child has ultrasound (US). Hydronephrosis needs a renogram **with a bladder catheter** and voiding cystography or videourodynamics.
- Check electrolytes, blood urea nitrogen (BUN) and creatinine. If raised, involve nephrology and measure 24-h urine **volume**: the reservoir must hold **3–4 h** of output.
- In **renal insufficiency**, acidosis **always worsens** with ileum or colon, so consider stomach. In **liver failure**, ammonia reabsorption is dangerous.
- Low-grade secondary **reflux** usually resolves after augmentation; correct high-grade reflux at reconstruction.
- In short gut, diarrhoea or faecal incontinence, spare the ileocaecal valve; add an antegrade continence enema (ACE) if needed.

- A sensate urethra, wheelchair use or prior bladder neck repair favour an abdominal stoma. Do all steps in **one operation** where possible.
- **Sterile urine** is mandatory; bowel preparation is optional. With a ventriculoperitoneal shunt, infection is rare but shunt failure increases.

3 · Managing the ureters

- Reimplant into the native bladder, with or without a **psoas hitch** (avoid the genitofemoral and femoral nerves). A short ureter needs transureteroureterostomy.
- Into bowel: **colon** under a taenia; **stomach** submucosally; **ileum** needs a **nipple valve** (hardest).

4 · Bladder augmentation (enterocystoplasty)

P1 Instestinal segments in urology — segments compared with electrolyte disturbance; NH_4Cl absorption mechanism; other metabolic, stone, cancer complications; renal function for continent diversion

Not in the paper answer:

- Base deficit is the most sensitive marker; height can fall even without frank acidosis.
- Mucus: colon more than ileum more than stomach; stomach forms uric acid stones.

Principles

- **Bivalve the native bladder widely**; a narrow anastomosis makes a diverticulum.
- **Always detubularise** and **reconfigure into a sphere**. Intact bowel contracts at **60–100 cm H2O**, and a sphere gives the most volume per surface area.
- Length: ileum **20–40 cm**, colon **~20 cm**; err on the large side.
- Use **no permanent sutures or staples**, because they seed stones.

Segments

- **Ileocystoplasty**: take ileum at least **15–20 cm proximal to ileocaecal valve** (to protect B12 absorption) and open it antimesenterically into a U or W.
- **Gastrocystoplasty** suits **renal insufficiency with acidosis, short gut and irradiated bowel**. **Jejunum** is avoided (salt-losing acidosis).

Segment	Advantages	Disadvantages
Ileum (default)	Least pressure activity; plentiful, mobile mesentery	Hyperchloraemic acidosis; B12 loss; mucus; hard to make antireflux
Ileocaecal	Taenial reimplant; appendix; imbricated valve	Diarrhoea in neurogenic bowel; more mucus
Sigmoid	Taenial reimplant; close to bladder	Strongest contractions; most mucus; more reaugmentation
Stomach	No acidosis; least mucus; fewer struvite stones; usable in renal failure	Hypokalaemic hypochloraemic alkalosis; haematuria–dysuria syndrome; tumours
Ureter	No mucus, no metabolic effect	Needs a massively dilated ureter of a non-functioning kidney; limited volume
Autoaugmentation (detrusorectomy)	Extraperitoneal; no bowel	Seldom adds capacity; compliance may not last

- Seromuscular and tissue-engineered augments are experimental. Technique matters more than segment, so use ileum unless there is a reason not to.

> **KEY POINT**
> Augmentation is needed less now, thanks to early anticholinergics, CIC, botulinum toxin and neuromodulation (immediate vs expectant treatment: 17% vs 41% augmented). This is not trial-proven, but it is the rationale for treating at diagnosis.

Incontinent alternatives

- **Vesicostomy** is an infant bridge for worsening hydronephrosis, febrile UTI with high-grade VUR, or high pressure. **Ileovesicostomy** suits those who cannot catheterise.

5 · Continent catheterisable channels

P2 Discuss principles of continence in continent urinary diversion. Application of Mitrofanoff principle in continent urinary and GI reconstruction. — flap vs nipple valve, conduit options (appendix, Monti, tapered ileum, ureter, Casale), appendicovesicostomy steps, MACE, results, stenosis

Not in the paper answer:

- In a **flap valve**, a narrow tube tunnelled into a **low-pressure** reservoir is **compressed** as pressure rises; the appendix needs only **~2 cm** of tunnel. Nipple valves (Kock, Mainz) are **effaced** by pressure (about 85% vs more than 90–95% continent). A flap valve has **no pop-off**.

> **ALGORITHM**
> 1. Detach the appendix with a caecal cuff and preserve the mesoappendix.
> 2. Tunnel the **distal end** submucosally.
> 3. Bring the caecal end to skin without tension or twist, with a **skin flap** in the stoma.
> 4. Keep the channel **short and straight**; fix the bladder to the abdominal wall.
> 5. Test catheterisation after each step.
> 6. Stent with 6–12 French (Fr) for 10–14 days; the surgeon does the first catheterisation.
> 7. Then CIC at least 4-hourly by day.

- Site the stoma at the umbilicus or right lower quadrant. The appendix can be **split** for both an ACE and a channel.
- **Yang–Monti**: a 1–2 cm ileal ring is retubularised **transversely** over 12 Fr; it needs about twice the revisions of the appendix. The **Casale** bladder flap spares bowel but has high stenosis.
- **Stomal stenosis** is the commonest complication and occurs early. Stomal incontinence (rare) means a short tunnel or high reservoir pressure.
- **Ureterosigmoidostomy and Mainz II** are largely abandoned because of acidosis, pyelonephritis and cancer.

6 · Bladder outlet procedures

Outlet procedures sacrifice voiding and pop-off, so neurogenic repairs need **concurrent augmentation** and surveillance for **≥5 yr**.

Procedure	Mechanism	Voiding	Main problem / place
Fascial sling	Coaptation and elevation	Rarely adequate, so CIC	Better in girls; only predictor of success is augmentation; AUS still possible later
Artificial urinary sphincter (AUS)	Inflatable bladder-neck cuff	**Preserved**	Infection, erosion, malfunction; best if voiding
Bulking injection	Submucosal	Preserved	Poor durability
Young–Dees–Leadbetter	Tubularised trigone	Possible	Exstrophy (see Ch 43); poor in neurogenic
Kropp	Anterior bladder tube under trigone (flap valve)	Abolished	Highest continence; hard urethral CIC (boys); new VUR
Pippi Salle	Anterior flap onlaid on trigonal strip	Abolished	Easier CIC; lower night continence; fistula
Bladder neck closure	Complete separation and continent stoma	Abolished	After failed repairs; no pop-off

- The **sling** suits a marginal bladder neck in an augmented child on CIC.
- The **AUS** cuff goes at the bladder neck in girls and prepubertal boys and is left **deactivated ~6 weeks**; most neurogenic children still need CIC. New hydronephrosis means unmasked bladder hostility.
- **Kropp** and **Salle** repairs often need later augmentation. Bladder neck **closure** with **interposed omentum** keeps fistula low, and suits reliable catheterisers only.

> **KEY POINT**
> Choosing the outlet: a child who voids gets an AUS (the only option preserving voiding). A child on CIC who is being augmented gets a sling (girls) or flap-valve lengthening with a channel. After failed repairs, do bladder neck closure with a Mitrofanoff. Always re-study the bladder afterwards.

7 · Malone antegrade continence enema (MACE)

- † An appendicocaecostomy gives scheduled antegrade washouts and faecal continence in spina bifida, anorectal malformation and refractory constipation. Plan it with the urinary reconstruction.

8 · Postoperative care

- Irrigate the suprapubic tube **≥3 times daily**. Do a **cystogram at ~3 weeks**, then start CIC 2–3-hourly by day, **never beyond 4–5 h**.

9 · Complications of augmentation

*Up to ⅓ need further surgery: **bowel is not bladder**.*

Segment	Derangement	Management
Ileum, colon	**Hyperchloraemic metabolic acidosis** (overt mainly with renal impairment)	Bicarbonate; **base deficit** is most sensitive
Jejunum	Hyponatraemic, hypochloraemic, hyperkalaemic acidosis	Avoid

Segment	Derangement	Management
Stomach	**Hypokalaemic hypochloraemic alkalosis** with vomiting; **haematuria–dysuria**	Fluids, H2 blockers or proton pump inhibitor
Stomach and bowel	Near neutral	Complex patients

- Acid load demineralises bone and lowers height, so treat base deficit early. Check B12 long term.
- **Bacteriuria** is expected with CIC; treat only symptomatic infection and **urea-splitters**.
- **Stones** are mostly **struvite**; prevent them with **daily irrigation ≥240 mL** saline, which also clears mucus.

Delayed spontaneous perforation

- Counsel a **~9%** risk, mostly in myelodysplasia. **Outlet procedures double the risk**, as do missed CIC and substance abuse.
- Signs are pain, distension, fever, sepsis and shoulder-tip pain; poor sensation means **late presentation** and deaths. Do a **cystogram** with a post-drain film, then a CT cystogram if negative. Treat by **immediate laparotomy and repair**; manage conservatively only if stable, sterile and contained.

Malignancy

P1 Cancer in augmented bladder — adenocarcinoma at anastomosis, earliest 4 yr, N-nitroso pathogenesis, gastrocystoplasty/exstrophy/transplant risk, SB 2–4%, nephrogenic adenoma, surveillance debate, cystectomy

- After ureterosigmoidostomy, adenocarcinoma arises at the anastomosis after **~26 yr**, from N-nitroso compounds.
- After augmentation it is reported from **4 yr**, is highest after **gastrocystoplasty**, and is aggressive. Bladder cancer in spina bifida is ~2–4%. There is no consensus on annual cystoscopy; consider it after gastrocystoplasty, in exstrophy and after transplant.

Pregnancy

- Vaginal delivery is possible; caesarean after major bladder neck repair or AUS, with a urologist present.

10 · Follow-up and transition

When	Tests
3, 6, 12 months	US; electrolytes, BUN, creatinine, urine culture (with or without urodynamics by 6 months)
Yearly, lifelong	US, chemistry (with or without blood gas, B12), CIC and irrigation compliance; urodynamics if new hydronephrosis, incontinence or VUR
Symptomatic	† Haematuria needs cystoscopy; pain or sepsis needs perforation excluded

- Late chronic kidney disease after augmentation is driven by **CIC non-compliance**, so adult urology must police CIC and the upper tracts.

11 · Landmark trials

Trial	Design	Result	Take-home
Early vs expectant neurogenic bladder care	Cohort, treated at diagnosis vs expectantly	Augmentation 17% vs 41%	Treat hostile bladder early
Daily bladder irrigation	Augmented cohort, ≥240 mL saline daily	Marked fall in stones	Routine daily irrigation
Tissue-engineered bladder	Industry multicentre trial	No gain in compliance or capacity	Bowel remains standard

★ Putting it together: common paediatric pathways

A · Antenatal urinary tract dilation (UTD)

ALGORITHM

1. **Prenatal:** A1 before 28 weeks: ultrasound (US) after 32 weeks. A2–3: US 4-weekly. Male, thick bladder, bilateral: suspect posterior urethral valves (PUV).
2. **Postnatal:** first US **after 48 hours**, except suspected PUV or bilateral severe dilation.
3. Normal (<10 mm): discharge; mild: repeat US; moderate–severe or dilated ureter: voiding cystourethrogram (VCUG), MAG3, consider prophylaxis (→ Ch 34).

B · First febrile urinary tract infection (UTI)

ALGORITHM

1. **Diagnose:** catheter or suprapubic sample if not toilet-trained; bag only rules out.
2. **Treat** within 48 h (>72 h doubles new scarring); oral if well.
3. **Image:** US all; VCUG if US abnormal, atypical or recurrent; DMSA at 4–6 months if needed.
4. **Prevent:** treat bladder and bowel dysfunction and constipation; prophylaxis if high risk (→ Ch 37, 41).

C · Newborn with PUV

ALGORITHM

1. Stabilise breathing → 5–7 Fr feeding tube → creatinine, electrolytes, fluids.
2. US, then VCUG; prophylaxis.
3. When stable, **endoscopic ablation**; if the urethra is small, vesicostomy.
4. Prognosis: first-year nadir creatinine (<0.8 mg/dL good); lifelong renal follow-up (→ Ch 45).

D · Neurogenic bladder in spina bifida

Step	What	Why
Birth	US, post-void residual, creatinine; latex-free care	Baseline
≈3 months	Urodynamics	Hostile: leak point or end-filling ≥40 cmH_2O, poor compliance, dyssynergia
Hostile bladder	Early clean intermittent catheterisation (CIC) and antimuscarinic	Kidney protection
Failing	Botulinum toxin, vesicostomy (infant), then augmentation	Safe storage (see Ch 46, 48)
Always	Bowel programme; watch for re-tethering	Safety

E · Daytime wetting and bedwetting

ALGORITHM

1. Urinalysis, residual with or without uroflow; exclude organic causes.
2. **Constipation first**, then urotherapy ≥4 weeks.
3. Antimuscarinic for overactive bladder, biofeedback for dysfunctional voiding; toxin or neuromodulation if refractory.
4. **Monosymptomatic enuresis:** alarm or desmopressin; combine if partial (→ Ch 47).

F · Augmentation segment

Segment	Advantages	Disadvantages
Ileum	Default; compliant	Mucus, hyperchloraemic acidosis, B12 loss
Ileocaecum / colon	Appendix for channel	Diarrhoea in neurogenic bowel; acidosis
Stomach	Suits renal failure; little mucus	Haematuria–dysuria, alkalosis
Ureter	No mucus or metabolic effect	Needs a massive ureter

KEY POINT

Reconstruction trades storage for emptying: an unreliably catheterised reservoir causes sepsis and perforation, so CIC commitment comes first.